PHARMACY MANAGEMENT SYSTEM

Object-Oriented Programming - Mini Project Report

Submitted in partial fulfillment of the requirements for the
Bachelor of Technology in Computer Science and Engineering

Project Title	Pharmacy Management System
Subject	Object-Oriented Programming (OOP)
Academic Year	2025 - 2026

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Introduction

The objective of this project is to design and implement a Pharmacy Management System (PMS) using Java and JavaFX to manage medicine inventory, customer records, billing transactions, supplier details, and prescription handling. This system enables pharmacy staff to register and manage medicines in stock, generate bills and invoices for customers, handle prescriptions provided by doctors, and receive automated alerts for low stock or expiring medicines. The system also focuses on ensuring secure data handling through role-based access control and persistent file-based storage using Java Serialization.

Problem Statement

Pharmacy Management System with Inventory Control and Billing Automation

Manual pharmacy operations are heavily dependent on paper-based records, leading to frequent billing errors, poor stock management, and difficulty in tracking medicine availability and expiry. Patients may receive incorrect medicines due to outdated records. There is no automated mechanism to alert staff about low stock or expired medicines. The Pharmacy Management System aims to automate and digitize all core pharmacy operations, improving accuracy, efficiency, and reliability.

Implementation Details

This project was implemented in the Java programming language using JavaFX for the graphical user interface. The application utilizes object-oriented programming principles to manage pharmacy operations effectively. The project contains 32 Java source files organized into four packages: model, service, ui, and util.

The complete list of source files is:

- Medicine.java
- MedicineCategory.java
- UserRole.java
- Customer.java
- Supplier.java
- Bill.java
- BillItem.java
- Prescription.java
- PrescriptionItem.java
- User.java
- InventoryManager.java
- BillingService.java
- CustomerService.java
- SupplierService.java
- PrescriptionService.java
- UserService.java
- ExpiryAlertThread.java
- StockAlertThread.java
- BillingThread.java

- BackupService.java
- PharmacyApp.java
- LoginScreen.java
- DashboardScreen.java
- InventoryScreen.java
- BillingScreen.java
- SupplierScreen.java
- CustomerScreen.java
- PrescriptionScreen.java
- ReportsScreen.java
- DataStore.java
- IDGenerator.java
- SampleDataLoader.java

Detailed Explanation of Each File

Model Package (com.pharmacy.model)

1. Medicine.java

Main model representing a medicine/drug item. Implements Serializable and Comparable<Medicine>. Uses Integer and Double wrapper classes with autoboxing/unboxing. Has constructor overloading (with and without supplierId) and a compareTo() method enabling Collections.sort() by medicine name.

Code for Medicine.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Medicine model class representing a medicine in the pharmacy.
* Demonstrates: Encapsulation, Serialization, Constructors, Wrapper classes.
*/
public class Medicine implements Serializable, Comparable<Medicine> {
private static final long serialVersionUID = 1L;
private String medicineId;
private String name;
private String manufacturer;
private MedicineCategory category;
private String batchNumber;
private String expiryDate; // Format: YYYY-MM-DD
private Integer quantity; // Wrapper class demonstration
private Double price; // Wrapper class demonstration
private String supplierId;
// Default constructor
public Medicine() {
}
// Parameterized constructor
public Medicine(String medicineId, String name, String manufacturer,
MedicineCategory category, String batchNumber,
String expiryDate, int quantity, double price, String supplierId) {
this.medicineId = medicineId;
```

```
this.name = name;
this.manufacturer = manufacturer;
this.category = category;
this.batchNumber = batchNumber;
this.expiryDate = expiryDate;
this.quantity = quantity; // Autoboxing: int -> Integer
this.price = price; // Autoboxing: double -> Double
this.supplierId = supplierId;
}
// Constructor overloading (without supplier)
public Medicine(String medicineId, String name, String manufacturer,
MedicineCategory category, String batchNumber,
String expiryDate, int quantity, double price) {
this(medicineId, name, manufacturer, category, batchNumber,
expiryDate, quantity, price, "");
}
// Getters and Setters (Encapsulation)
public String getMedicineId() { return medicineId; }
public void setMedicineId(String medicineId) { this.medicineId = medicineId; }
public String getName() { return name; }
public void setName(String name) { this.name = name; }
public String getManufacturer() { return manufacturer; }
public void setManufacturer(String manufacturer) { this.manufacturer = manufacturer; }
public MedicineCategory getCategory() { return category; }
public void setCategory(MedicineCategory category) { this.category = category; }
public String getBatchNumber() { return batchNumber; }
public void setBatchNumber(String batchNumber) { this.batchNumber = batchNumber; }
public String getExpiryDate() { return expiryDate; }
public void setExpiryDate(String expiryDate) { this.expiryDate = expiryDate; }
public Integer getQuantity() { return quantity; }
public void setQuantity(Integer quantity) { this.quantity = quantity; }
public Double getPrice() { return price; }
public void setPrice(Double price) { this.price = price; }
public String getSupplierId() { return supplierId; }
public void setSupplierId(String supplierId) { this.supplierId = supplierId; }
// Unboxing demonstration
public int getQuantityPrimitive() {
return quantity; // Unboxing: Integer -> int
}
public double getPricePrimitive() {
return price; // Unboxing: Double -> double
}
// Comparable implementation for sorting
@Override
public int compareTo(Medicine other) {
return this.name.compareToIgnoreCase(other.name);
}
@Override
public String toString() {
return medicineId + " | " + name + " | " + category + " | Qty: " + quantity
+ " | Rs." + String.format("%.2f", price) + " | Exp: " + expiryDate;
}
}
```

2. MedicineCategory.java

Enum defining the available medicine categories: TABLET, CAPSULE, SYRUP, INJECTION, OINTMENT, DROPS, INHALER, and POWDER. Demonstrates the use of Java enumerations.

Code for MedicineCategory.java:

```
package com.pharmacy.model;
/**
* Enum representing different categories of medicines.
* Demonstrates: Enum usage as required by OSDL syllabus.
*/
public enum MedicineCategory {
TABLET("Tablet"),
CAPSULE("Capsule"),
SYRUP("Syrup"),
INJECTION("Injection"),
OINTMENT("Ointment"),
DROPS("Drops"),
INHALER("Inhaler"),
POWDER("Powder");
private final String displayName;
// Enum constructor
MedicineCategory(String displayName) {
this.displayName = displayName;
}
public String getDisplayName() {
return displayName;
}
@Override
public String toString() {
return displayName;
}
}
```

3. UserRole.java

Enum defining user roles in the system: ADMIN and PHARMACIST. Used for role-based access control throughout the application.

Code for UserRole.java:

```
package com.pharmacy.model;
/**
* Enum for role-based access control.
* Demonstrates: Enum with methods.
*/
public enum UserRole {
ADMIN("Administrator", true),
PHARMACIST("Pharmacist", false);
private final String displayName;
private final boolean canManageUsers;
UserRole(String displayName, boolean canManageUsers) {
this.displayName = displayName;
this.canManageUsers = canManageUsers;
}
public String getDisplayName() {
return displayName;
```

```
}
public boolean canManageUsers() {
return canManageUsers;
}
}
```

4. Customer.java

Model class representing a pharmacy customer. Implements Serializable for persistent storage. Contains customerId, name, phone, email, and address fields with getters/setters.

Code for Customer.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Customer/Patient model class.
* Demonstrates: Encapsulation, Serialization.
*/
public class Customer implements Serializable {
private static final long serialVersionUID = 1L;
private String customerId;
private String name;
private String phone;
private String email;
private String address;
// Default constructor
public Customer() {
}
// Parameterized constructor
public Customer(String customerId, String name, String phone, String email, String address) {
this.customerId = customerId;
this.name = name;
this.phone = phone;
this.email = email;
this.address = address;
}
// Getters and Setters
public String getCustomerId() { return customerId; }
public void setCustomerId(String customerId) { this.customerId = customerId; }
public String getName() { return name; }
public void setName(String name) { this.name = name; }
public String getPhone() { return phone; }
public void setPhone(String phone) { this.phone = phone; }
public String getEmail() { return email; }
public void setEmail(String email) { this.email = email; }
public String getAddress() { return address; }
public void setAddress(String address) { this.address = address; }
@Override
public String toString() {
return customerId + " | " + name + " | " + phone;
}
}
```

5. Supplier.java

Model class representing a medicine supplier. Implements Serializable. Contains supplierId, companyName, contactPerson, phone, email, and address fields.

Code for Supplier.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Supplier model class.
* Demonstrates: Encapsulation, Serialization.
*/
public class Supplier implements Serializable {
private static final long serialVersionUID = 1L;
private String supplierId;
private String companyName;
private String contactPerson;
private String phone;
private String email;
private String address;
// Default constructor
public Supplier() {
}
// Parameterized constructor
public Supplier(String supplierId, String companyName, String contactPerson,
String phone, String email, String address) {
this.supplierId = supplierId;
this.companyName = companyName;
this.contactPerson = contactPerson;
this.phone = phone;
this.email = email;
this.address = address;
}
// Getters and Setters
public String getSupplierId() { return supplierId; }
public void setSupplierId(String supplierId) { this.supplierId = supplierId; }
public String getCompanyName() { return companyName; }
public void setCompanyName(String companyName) { this.companyName = companyName; }
public String getContactPerson() { return contactPerson; }
public void setContactPerson(String contactPerson) { this.contactPerson = contactPerson; }
public String getPhone() { return phone; }
public void setPhone(String phone) { this.phone = phone; }
public String getEmail() { return email; }
public void setEmail(String email) { this.email = email; }
public String getAddress() { return address; }
public void setAddress(String address) { this.address = address; }
@Override
public String toString() {
return supplierId + " | " + companyName + " | " + contactPerson;
}
}
```

6. Bill.java

Model class representing a billing transaction. Contains overloaded applyDiscount() methods demonstrating method overloading. Stores bill items, total amount, and date.

Code for Bill.java:

```
package com.pharmacy.model;
import java.io.Serializable;
import java.util.ArrayList;
/**
* Bill/Invoice model class for sales transactions.
* Demonstrates: Encapsulation, Serialization, ArrayList, Wrapper classes.
*/
public class Bill implements Serializable {
private static final long serialVersionUID = 1L;
private String billId;
private String customerId;
private String customerName;
private String date;
private ArrayList<BillItem> items;
private Double totalAmount; // Wrapper class
private Double discount;
private Double netAmount;
private String billedBy; // username of the person who billed
// Default constructor
public Bill() {
this.items = new ArrayList<>();
this.totalAmount = 0.0;
this.discount = 0.0;
this.netAmount = 0.0;
}
// Parameterized constructor
public Bill(String billId, String customerId, String customerName,
String date, String billedBy) {
this.billId = billId;
this.customerId = customerId;
this.customerName = customerName;
this.date = date;
this.billedBy = billedBy;
this.items = new ArrayList<>();
this.totalAmount = 0.0;
this.discount = 0.0;
this.netAmount = 0.0;
}
// Add item and recalculate total
public void addItem(BillItem item) {
items.add(item);
recalculate();
}
// Recalculate totals
public void recalculate() {
double total = 0.0;
for (BillItem item : items) {
total += item.getSubtotal(); // Unboxing
}
this.totalAmount = total; // Autoboxing
this.netAmount = totalAmount - discount;
}
// Method overloading: apply discount by percentage
public void applyDiscount(double percentage) {
```

```
this.discount = totalAmount * (percentage / 100.0);
this.netAmount = totalAmount - discount;
}
// Method overloading: apply discount by fixed amount with a flag
public void applyDiscount(double amount, boolean isFixed) {
if (isFixed) {
this.discount = amount;
} else {
this.discount = totalAmount * (amount / 100.0);
}
this.netAmount = totalAmount - discount;
}
// Getters and Setters
public String getBillId() { return billId; }
public void setBillId(String billId) { this.billId = billId; }
public String getCustomerId() { return customerId; }
public void setCustomerId(String customerId) { this.customerId = customerId; }
public String getCustomerName() { return customerName; }
public void setCustomerName(String customerName) { this.customerName = customerName; }
public String getDate() { return date; }
public void setDate(String date) { this.date = date; }
public ArrayList<BillItem> getItems() { return items; }
public void setItems(ArrayList<BillItem> items) { this.items = items; }
public Double getTotalAmount() { return totalAmount; }
public void setTotalAmount(Double totalAmount) { this.totalAmount = totalAmount; }
public Double getDiscount() { return discount; }
public void setDiscount(Double discount) { this.discount = discount; }
public Double getNetAmount() { return netAmount; }
public void setNetAmount(Double netAmount) { this.netAmount = netAmount; }
public String getBilledBy() { return billedBy; }
public void setBilledBy(String billedBy) { this.billedBy = billedBy; }
@Override
public String toString() {
return billId + " | " + customerName + " | Rs." + String.format("%.2f", netAmount)
+ " | " + date;
}
}
```

7. BillItem.java

Represents a single line item in a bill. Uses Integer for quantity and Double for unitPrice and subtotal, demonstrating wrapper classes and autoboxing.

Code for BillItem.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Individual item in a bill.
* Demonstrates: Encapsulation, Serialization, Wrapper classes.
*/
public class BillItem implements Serializable {
private static final long serialVersionUID = 1L;
private String medicineId;
private String medicineName;
private Integer quantity; // Wrapper class
```

```
private Double unitPrice; // Wrapper class
private Double subtotal; // Wrapper class
public BillItem() {
}
public BillItem(String medicineId, String medicineName, int quantity, double unitPrice) {
this.medicineId = medicineId;
this.medicineName = medicineName;
this.quantity = quantity; // Autoboxing
this.unitPrice = unitPrice; // Autoboxing
this.subtotal = quantity * unitPrice; // Autoboxing result
}
// Getters and Setters
public String getMedicineId() { return medicineId; }
public void setMedicineId(String medicineId) { this.medicineId = medicineId; }
public String getMedicineName() { return medicineName; }
public void setMedicineName(String medicineName) { this.medicineName = medicineName; }
public Integer getQuantity() { return quantity; }
public void setQuantity(Integer quantity) { this.quantity = quantity; }
public Double getUnitPrice() { return unitPrice; }
public void setUnitPrice(Double unitPrice) { this.unitPrice = unitPrice; }
public Double getSubtotal() { return subtotal; }
public void setSubtotal(Double subtotal) { this.subtotal = subtotal; }
@Override
public String toString() {
return medicineName + " x" + quantity + " @ Rs." + String.format("%.2f", unitPrice)
+ " = Rs." + String.format("%.2f", subtotal);
}
}
```

8. Prescription.java

Represents a medical prescription issued to a customer. Contains a list of PrescriptionItem objects and links to the customer and prescribing doctor.

Code for Prescription.java:

```
package com.pharmacy.model;
import java.io.Serializable;
import java.util.ArrayList;
/**
* Prescription model class linking customer to prescribed medicines.
* Demonstrates: Encapsulation, Serialization, ArrayList usage.
*/
public class Prescription implements Serializable {
private static final long serialVersionUID = 1L;
private String prescriptionId;
private String customerId;
private String customerName;
private String doctorName;
private String date;
private ArrayList<PrescriptionItem> items;
private String notes;
// Default constructor
public Prescription() {
this.items = new ArrayList<>();
}
```

```
// Parameterized constructor
public Prescription(String prescriptionId, String customerId, String customerName,
String doctorName, String date, String notes) {
this.prescriptionId = prescriptionId;
this.customerId = customerId;
this.customerName = customerName;
this.doctorName = doctorName;
this.date = date;
this.notes = notes;
this.items = new ArrayList<>();
}
// Add item to prescription
public void addItem(PrescriptionItem item) {
items.add(item);
}
// Getters and Setters
public String getPrescriptionId() { return prescriptionId; }
public void setPrescriptionId(String prescriptionId) { this.prescriptionId = prescriptionId; }
public String getCustomerId() { return customerId; }
public void setCustomerId(String customerId) { this.customerId = customerId; }
public String getCustomerName() { return customerName; }
public void setCustomerName(String customerName) { this.customerName = customerName; }
public String getDoctorName() { return doctorName; }
public void setDoctorName(String doctorName) { this.doctorName = doctorName; }
public String getDate() { return date; }
public void setDate(String date) { this.date = date; }
public ArrayList<PrescriptionItem> getItems() { return items; }
public void setItems(ArrayList<PrescriptionItem> items) { this.items = items; }
public String getNotes() { return notes; }
public void setNotes(String notes) { this.notes = notes; }
@Override
public String toString() {
return prescriptionId + " | Patient: " + customerName + " | Dr. " + doctorName + " | " + date;
}
}
```

9. PrescriptionItem.java

Represents a single medicine entry within a prescription, including medicine name, dosage, and quantity instructions.

Code for PrescriptionItem.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Individual item within a prescription.
* Demonstrates: Encapsulation, Serialization.
*/
public class PrescriptionItem implements Serializable {
private static final long serialVersionUID = 1L;
private String medicineId;
private String medicineName;
private Integer quantity;
private String dosage;
public PrescriptionItem() {
```

```
}
public PrescriptionItem(String medicineId, String medicineName, int quantity, String dosage) {
this.medicineId = medicineId;
this.medicineName = medicineName;
this.quantity = quantity; // Autoboxing
this.dosage = dosage;
}
public String getMedicineId() { return medicineId; }
public void setMedicineId(String medicineId) { this.medicineId = medicineId; }
public String getMedicineName() { return medicineName; }
public void setMedicineName(String medicineName) { this.medicineName = medicineName; }
public Integer getQuantity() { return quantity; }
public void setQuantity(Integer quantity) { this.quantity = quantity; }
public String getDosage() { return dosage; }
public void setDosage(String dosage) { this.dosage = dosage; }
@Override
public String toString() {
return medicineName + " x" + quantity + " (" + dosage + ")";
}
}
```

10. User.java

Represents a system user with username, password, fullName, and UserRole. Contains an authenticate() method that validates login credentials.

Code for User.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* User model for login and role-based access.
* Demonstrates: Encapsulation, Enum usage, Serialization.
*/
public class User implements Serializable {
private static final long serialVersionUID = 1L;
private String username;
private String password;
private String fullName;
private UserRole role;
// Default constructor
public User() {
}
// Parameterized constructor
public User(String username, String password, String fullName, UserRole role) {
this.username = username;
this.password = password;
this.fullName = fullName;
this.role = role;
}
// Validate login credentials
public boolean authenticate(String inputPassword) {
return this.password.equals(inputPassword);
}
// Getters and Setters
public String getUsername() { return username; }
```

```
public void setUsername(String username) { this.username = username; }
public String getPassword() { return password; }
public void setPassword(String password) { this.password = password; }
public String getFullName() { return fullName; }
public void setFullName(String fullName) { this.fullName = fullName; }
public UserRole getRole() { return role; }
public void setRole(UserRole role) { this.role = role; }
@Override
public String toString() {
return username + " (" + role.getDisplayName() + ")";
}
}
```

Service Package (com.pharmacy.service)

11. InventoryManager.java

Core service class managing the medicine inventory. Uses ArrayList for ordered storage and HashMap for fast lookup. All write methods are synchronized for thread safety. Uses Iterator for safe deletion, and Collections.sort() with both Comparable and Comparator for flexible sorting.

Code for InventoryManager.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Medicine;
import com.pharmacy.model.MedicineCategory;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;
import java.util.HashMap;
import java.util.Iterator;
/**
* Manages the medicine inventory.
* Demonstrates: ArrayList, HashMap, Iterator, Collections.sort, Generics, Synchronization.
*/
public class InventoryManager {
private ArrayList<Medicine> medicineList;
private HashMap<String, Medicine> medicineMap; // Maps medicineId -> Medicine object
private DataStore<Medicine> dataStore;
private static final String DATA_FILE = "data/medicines.dat";
private static final String BACKUP_FILE = "data/backup_medicines.txt";
private static final int LOW_STOCK_THRESHOLD = 10;
public InventoryManager() {
this.dataStore = new DataStore<>(DATA_FILE);
this.medicineList = dataStore.loadAll();
this.medicineMap = new HashMap<>();
// Populate HashMap from loaded list
for (Medicine med : medicineList) {
medicineMap.put(med.getMedicineId(), med);
}
}
// Add medicine (synchronized for thread safety)
public synchronized void addMedicine(Medicine medicine) {
medicineList.add(medicine);
medicineMap.put(medicine.getMedicineId(), medicine);
```

```
saveData();
}
// Update medicine
public synchronized void updateMedicine(Medicine medicine) {
medicineMap.put(medicine.getMedicineId(), medicine);
// Update in list as well
for (int i = 0; i < medicineList.size(); i++) {
if (medicineList.get(i).getMedicineId().equals(medicine.getMedicineId())) {
medicineList.set(i, medicine);
break;
}
}
saveData();
}
// Delete medicine using Iterator
public synchronized void deleteMedicine(String medicineId) {
Iterator<Medicine> iterator = medicineList.iterator();
while (iterator.hasNext()) {
Medicine med = iterator.next();
if (med.getMedicineId().equals(medicineId)) {
iterator.remove();
break;
}
}
medicineMap.remove(medicineId);
saveData();
}
// Get medicine by ID using HashMap
public Medicine getMedicineById(String medicineId) {
return medicineMap.get(medicineId);
}
// Search medicines by name
public ArrayList<Medicine> searchByName(String name) {
ArrayList<Medicine> results = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getName().toLowerCase().contains(name.toLowerCase())) {
results.add(med);
}
}
return results;
}
// Filter by category
public ArrayList<Medicine> filterByCategory(MedicineCategory category) {
ArrayList<Medicine> results = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getCategory() == category) {
results.add(med);
}
}
return results;
}
// Sort by name using Collections.sort (Comparable)
public ArrayList<Medicine> getSortedByName() {
ArrayList<Medicine> sorted = new ArrayList<>(medicineList);
Collections.sort(sorted); // Uses Medicine's compareTo
return sorted;
```

```
}
// Sort by price using Comparator
public ArrayList<Medicine> getSortedByPrice() {
ArrayList<Medicine> sorted = new ArrayList<>(medicineList);
Collections.sort(sorted, new Comparator<Medicine>() {
@Override
public int compare(Medicine m1, Medicine m2) {
return Double.compare(m1.getPrice(), m2.getPrice());
}
});
return sorted;
}
// Sort by expiry date
public ArrayList<Medicine> getSortedByExpiry() {
ArrayList<Medicine> sorted = new ArrayList<>(medicineList);
Collections.sort(sorted, new Comparator<Medicine>() {
@Override
public int compare(Medicine m1, Medicine m2) {
return m1.getExpiryDate().compareTo(m2.getExpiryDate());
}
});
return sorted;
}
// Get low stock medicines
public ArrayList<Medicine> getLowStockMedicines() {
ArrayList<Medicine> lowStock = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getQuantity() < LOW_STOCK_THRESHOLD) {
lowStock.add(med);
}
}
return lowStock;
}
// Get expired medicines (compare with current date string YYYY-MM-DD)
public ArrayList<Medicine> getExpiredMedicines(String currentDate) {
ArrayList<Medicine> expired = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getExpiryDate().compareTo(currentDate) <= 0) {
expired.add(med);
}
}
return expired;
}
// Get medicines expiring soon (within 30 days approximation)
public ArrayList<Medicine> getExpiringSoon(String currentDate, String thresholdDate) {
ArrayList<Medicine> expiring = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getExpiryDate().compareTo(currentDate) > 0
&& med.getExpiryDate().compareTo(thresholdDate) <= 0) {
expiring.add(med);
}
}
return expiring;
}
// Reduce stock (for billing)
public synchronized boolean reduceStock(String medicineId, int quantity) {
```

```
Medicine med = medicineMap.get(medicineId);
if (med != null && med.getQuantity() >= quantity) {
med.setQuantity(med.getQuantity() - quantity);
saveData();
return true;
}
return false;
}
// Get all medicines
public ArrayList<Medicine> getAllMedicines() {
return medicineList;
}
// Save data
private void saveData() {
dataStore.saveAll(medicineList);
}
// Create backup
public void createBackup() {
dataStore.backupToText(medicineList, BACKUP_FILE);
}
public int getMedicineCount() {
return medicineList.size();
}
}
```

12. BillingService.java

Handles all billing operations: creating bills, adding items, calculating totals, applying discounts, and generating invoice text files using FileWriter. All methods are synchronized.

Code for BillingService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Bill;
import com.pharmacy.model.BillItem;
import com.pharmacy.util.DataStore;
import java.io.FileWriter;
import java.io.IOException;
import java.util.ArrayList;
/**
* Manages billing and invoice operations.
* Demonstrates: ArrayList, Serialization, FileWriter, Method overloading.
*/
public class BillingService {
private ArrayList<Bill> billList;
private DataStore<Bill> dataStore;
private InventoryManager inventoryManager;
private static final String DATA_FILE = "data/bills.dat";
private static final String BACKUP_FILE = "data/backup_bills.txt";
private static final String INVOICE_DIR = "data/invoices/";
public BillingService(InventoryManager inventoryManager) {
this.inventoryManager = inventoryManager;
this.dataStore = new DataStore<>(DATA_FILE);
this.billList = dataStore.loadAll();
}
// Create and save a bill
```

```
public synchronized boolean createBill(Bill bill) {
// Reduce stock for each item
for (BillItem item : bill.getItems()) {
boolean success = inventoryManager.reduceStock(item.getMedicineId(), item.getQuantity());
if (!success) {
return false; // Insufficient stock
}
}
billList.add(bill);
saveData();
generateInvoice(bill);
return true;
}
/**
* Generate invoice as text file.
* Demonstrates: FileWriter usage.
*/
public void generateInvoice(Bill bill) {
java.io.File dir = new java.io.File(INVOICE_DIR);
if (!dir.exists()) {
dir.mkdirs();
}
String invoicePath = INVOICE_DIR + bill.getBillId() + "_invoice.txt";
try (FileWriter writer = new FileWriter(invoicePath)) {
writer.write("■■■
```

```
}
// Get bills by date
public ArrayList<Bill> getBillsByDate(String date) {
ArrayList<Bill> results = new ArrayList<>();
for (Bill bill : billList) {
if (bill.getDate().equals(date)) {
results.add(bill);
}
}
return results;
}
// Get total sales
public double getTotalSales() {
double total = 0.0;
for (Bill bill : billList) {
total += bill.getNetAmount();
}
return total;
}
// Get total sales for a date
public double getTotalSalesByDate(String date) {
double total = 0.0;
for (Bill bill : billList) {
if (bill.getDate().equals(date)) {
total += bill.getNetAmount();
}
}
return total;
}
private void saveData() {
dataStore.saveAll(billList);
}
public void createBackup() {
dataStore.backupToText(billList, BACKUP_FILE);
}
public int getBillCount() {
return billList.size();
}
}
```

13. CustomerService.java

Provides CRUD operations for customer records using ArrayList and HashMap. Uses Iterator for safe removal. Persists data using DataStore serialization.

Code for CustomerService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Customer;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Iterator;
/**
* Manages customer/patient records.
* Demonstrates: ArrayList, HashMap, Iterator, Generics.
*/
```

```
public class CustomerService {
private ArrayList<Customer> customerList;
private HashMap<String, Customer> customerMap;
private DataStore<Customer> dataStore;
private static final String DATA_FILE = "data/customers.dat";
private static final String BACKUP_FILE = "data/backup_customers.txt";
public CustomerService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.customerList = dataStore.loadAll();
this.customerMap = new HashMap<>();
for (Customer cust : customerList) {
customerMap.put(cust.getCustomerId(), cust);
}
}
public synchronized void addCustomer(Customer customer) {
customerList.add(customer);
customerMap.put(customer.getCustomerId(), customer);
saveData();
}
public synchronized void updateCustomer(Customer customer) {
customerMap.put(customer.getCustomerId(), customer);
for (int i = 0; i < customerList.size(); i++) {
if (customerList.get(i).getCustomerId().equals(customer.getCustomerId())) {
customerList.set(i, customer);
break;
}
}
saveData();
}
public synchronized void deleteCustomer(String customerId) {
Iterator<Customer> iterator = customerList.iterator();
while (iterator.hasNext()) {
Customer cust = iterator.next();
if (cust.getCustomerId().equals(customerId)) {
iterator.remove();
break;
}
}
customerMap.remove(customerId);
saveData();
}
public Customer getCustomerById(String customerId) {
return customerMap.get(customerId);
}
public ArrayList<Customer> searchByName(String name) {
ArrayList<Customer> results = new ArrayList<>();
for (Customer cust : customerList) {
if (cust.getName().toLowerCase().contains(name.toLowerCase())) {
results.add(cust);
}
}
return results;
}
public ArrayList<Customer> getAllCustomers() {
return customerList;
```

```
}
private void saveData() {
dataStore.saveAll(customerList);
}
public void createBackup() {
dataStore.backupToText(customerList, BACKUP_FILE);
}
public int getCustomerCount() {
return customerList.size();
}
}
```

14. SupplierService.java

Provides CRUD operations for supplier records using ArrayList and HashMap. Uses Iterator for safe removal. Persists data using DataStore serialization.

Code for SupplierService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Supplier;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Iterator;
/**
* Manages supplier records.
* Demonstrates: ArrayList, HashMap, Iterator, Generics.
*/
public class SupplierService {
private ArrayList<Supplier> supplierList;
private HashMap<String, Supplier> supplierMap;
private DataStore<Supplier> dataStore;
private static final String DATA_FILE = "data/suppliers.dat";
private static final String BACKUP_FILE = "data/backup_suppliers.txt";
public SupplierService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.supplierList = dataStore.loadAll();
this.supplierMap = new HashMap<>();
for (Supplier sup : supplierList) {
supplierMap.put(sup.getSupplierId(), sup);
}
}
public synchronized void addSupplier(Supplier supplier) {
supplierList.add(supplier);
supplierMap.put(supplier.getSupplierId(), supplier);
saveData();
}
public synchronized void updateSupplier(Supplier supplier) {
supplierMap.put(supplier.getSupplierId(), supplier);
for (int i = 0; i < supplierList.size(); i++) {
if (supplierList.get(i).getSupplierId().equals(supplier.getSupplierId())) {
supplierList.set(i, supplier);
break;
}
}
```

```
saveData();
}
public synchronized void deleteSupplier(String supplierId) {
Iterator<Supplier> iterator = supplierList.iterator();
while (iterator.hasNext()) {
Supplier sup = iterator.next();
if (sup.getSupplierId().equals(supplierId)) {
iterator.remove();
break;
}
}
supplierMap.remove(supplierId);
saveData();
}
public Supplier getSupplierById(String supplierId) {
return supplierMap.get(supplierId);
}
public ArrayList<Supplier> searchByName(String name) {
ArrayList<Supplier> results = new ArrayList<>();
for (Supplier sup : supplierList) {
if (sup.getCompanyName().toLowerCase().contains(name.toLowerCase())) {
results.add(sup);
}
}
return results;
}
public ArrayList<Supplier> getAllSuppliers() {
return supplierList;
}
private void saveData() {
dataStore.saveAll(supplierList);
}
public void createBackup() {
dataStore.backupToText(supplierList, BACKUP_FILE);
}
public int getSupplierCount() {
return supplierList.size();
}
}
```

15. PrescriptionService.java

Manages prescription records with full CRUD support. Uses ArrayList and HashMap, Iterator for deletion, and DataStore for serialized persistence.

Code for PrescriptionService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Prescription;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.HashMap;
/**
* Manages prescription records.
* Demonstrates: ArrayList, HashMap, Generics, Serialization.
*/
```

```
public class PrescriptionService {
private ArrayList<Prescription> prescriptionList;
private HashMap<String, Prescription> prescriptionMap;
private DataStore<Prescription> dataStore;
private static final String DATA_FILE = "data/prescriptions.dat";
private static final String BACKUP_FILE = "data/backup_prescriptions.txt";
public PrescriptionService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.prescriptionList = dataStore.loadAll();
this.prescriptionMap = new HashMap<>();
for (Prescription pres : prescriptionList) {
prescriptionMap.put(pres.getPrescriptionId(), pres);
}
}
public synchronized void addPrescription(Prescription prescription) {
prescriptionList.add(prescription);
prescriptionMap.put(prescription.getPrescriptionId(), prescription);
saveData();
}
public synchronized void deletePrescription(String prescriptionId) {
prescriptionList.removeIf(p -> p.getPrescriptionId().equals(prescriptionId));
prescriptionMap.remove(prescriptionId);
saveData();
}
public Prescription getPrescriptionById(String prescriptionId) {
return prescriptionMap.get(prescriptionId);
}
public ArrayList<Prescription> getByCustomerId(String customerId) {
ArrayList<Prescription> results = new ArrayList<>();
for (Prescription pres : prescriptionList) {
if (pres.getCustomerId().equals(customerId)) {
results.add(pres);
}
}
return results;
}
public ArrayList<Prescription> getAllPrescriptions() {
return prescriptionList;
}
private void saveData() {
dataStore.saveAll(prescriptionList);
}
public void createBackup() {
dataStore.backupToText(prescriptionList, BACKUP_FILE);
}
public int getPrescriptionCount() {
return prescriptionList.size();
}
}
```

16. UserService.java

Handles user authentication and management. Creates default admin and pharmacist accounts on first run. Validates credentials using User.authenticate().

Code for UserService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.User;
import com.pharmacy.model.UserRole;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
/**
* Manages user authentication and accounts.
* Demonstrates: ArrayList, Serialization, role-based access.
*/
public class UserService {
private ArrayList<User> userList;
private DataStore<User> dataStore;
private User currentUser;
private static final String DATA_FILE = "data/users.dat";
public UserService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.userList = dataStore.loadAll();
// Create default admin if no users exist
if (userList.isEmpty()) {
createDefaultUsers();
}
}
// Create default users for first run
private void createDefaultUsers() {
userList.add(new User("admin", "admin123", "System Administrator", UserRole.ADMIN));
userList.add(new User("pharmacist", "pharma123", "Default Pharmacist", UserRole.PHARMACIST));
dataStore.saveAll(userList);
}
// Authenticate user
public User authenticate(String username, String password) {
for (User user : userList) {
if (user.getUsername().equals(username) && user.authenticate(password)) {
this.currentUser = user;
return user;
}
}
return null;
}
public User getCurrentUser() {
return currentUser;
}
public void setCurrentUser(User user) {
this.currentUser = user;
}
public void addUser(User user) {
userList.add(user);
dataStore.saveAll(userList);
}
public ArrayList<User> getAllUsers() {
return userList;
}
public void logout() {
this.currentUser = null;
```

```
}
}
```

17. ExpiryAlertThread.java

Background thread that monitors medicines approaching their expiry date. Extends the Thread class directly, runs as a daemon thread, uses sleep(30000) to check every 30 seconds, and uses wait()/notify() for synchronization.

Code for ExpiryAlertThread.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Medicine;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.Date;
/**
* Background thread that checks for expired and soon-to-expire medicines.
* Demonstrates: Thread class, sleep(), join(), synchronized, Runnable.
*/
public class ExpiryAlertThread extends Thread {
private InventoryManager inventoryManager;
private volatile boolean running = true;
private ArrayList<Medicine> expiredMedicines;
private ArrayList<Medicine> expiringMedicines;
private final Object lock = new Object();
private AlertListener listener;
// Interface for callback (Abstraction demonstration)
public interface AlertListener {
void onAlertUpdate(ArrayList<Medicine> expired, ArrayList<Medicine> expiringSoon);
}
public ExpiryAlertThread(InventoryManager inventoryManager) {
super("ExpiryAlertThread");
this.inventoryManager = inventoryManager;
this.expiredMedicines = new ArrayList<>();
this.expiringMedicines = new ArrayList<>();
setDaemon(true); // Daemon thread - will stop when main thread stops
}
public void setAlertListener(AlertListener listener) {
this.listener = listener;
}
@Override
public void run() {
while (running) {
try {
checkExpiry();
// Sleep for 30 seconds between checks
Thread.sleep(30000);
} catch (InterruptedException e) {
Thread.currentThread().interrupt();
break;
}
}
}
// Synchronized method for thread-safe expiry checking
```

```
private synchronized void checkExpiry() {
SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");
String currentDate = sdf.format(new Date());
// Calculate date 30 days from now
Calendar cal = Calendar.getInstance();
cal.add(Calendar.DAY_OF_MONTH, 30);
String thresholdDate = sdf.format(cal.getTime());
synchronized (lock) {
expiredMedicines = inventoryManager.getExpiredMedicines(currentDate);
expiringMedicines = inventoryManager.getExpiringSoon(currentDate, thresholdDate);
// Notify waiting threads using wait/notify mechanism
lock.notifyAll();
}
// Callback to listener
if (listener != null) {
listener.onAlertUpdate(expiredMedicines, expiringMedicines);
}
}
// Method for other threads to wait for alert data
public ArrayList<Medicine> waitForExpiredList() throws InterruptedException {
synchronized (lock) {
lock.wait(5000); // Wait up to 5 seconds
return new ArrayList<>(expiredMedicines);
}
}
public ArrayList<Medicine> getExpiredMedicines() {
synchronized (lock) {
return new ArrayList<>(expiredMedicines);
}
}
public ArrayList<Medicine> getExpiringMedicines() {
synchronized (lock) {
return new ArrayList<>(expiringMedicines);
}
}
public void stopChecking() {
running = false;
this.interrupt();
}
}
```

18. StockAlertThread.java

Background thread that monitors low stock levels. Implements the Runnable interface (second threading approach). Uses a volatile boolean flag for graceful shutdown and a synchronized checkStock() method.

Code for StockAlertThread.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Medicine;
import java.util.ArrayList;
/**
* Background thread that monitors low stock levels.
* Demonstrates: Runnable interface, Thread, sleep(), synchronized.
*/
```

```
public class StockAlertThread implements Runnable {
private InventoryManager inventoryManager;
private volatile boolean running = true;
private ArrayList<Medicine> lowStockMedicines;
private AlertListener listener;
// Callback interface (Abstraction)
public interface AlertListener {
void onLowStockUpdate(ArrayList<Medicine> lowStockMedicines);
}
public StockAlertThread(InventoryManager inventoryManager) {
this.inventoryManager = inventoryManager;
this.lowStockMedicines = new ArrayList<>();
}
public void setAlertListener(AlertListener listener) {
this.listener = listener;
}
@Override
public void run() {
while (running) {
try {
checkStock();
Thread.sleep(30000); // Check every 30 seconds
} catch (InterruptedException e) {
Thread.currentThread().interrupt();
break;
}
}
}
private synchronized void checkStock() {
lowStockMedicines = inventoryManager.getLowStockMedicines();
if (listener != null) {
listener.onLowStockUpdate(lowStockMedicines);
}
}
public synchronized ArrayList<Medicine> getLowStockMedicines() {
return new ArrayList<>(lowStockMedicines);
}
public void stopChecking() {
running = false;
}
}
```

19. BillingThread.java

Processes billing operations asynchronously. Extends Thread and uses sleep() and join(). Defines a BillingCallback inner interface for returning results to the UI thread via Platform.runLater().

Code for BillingThread.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Bill;
/**
* Multi-threaded billing simulation.
* Demonstrates: Thread, Runnable, synchronized, sleep(), join().
*
* Simulates billing processing in a separate thread to keep UI responsive.
```

```
*/
public class BillingThread extends Thread {
private BillingService billingService;
private Bill bill;
private boolean success;
private String message;
private BillingCallback callback;
// Callback interface for billing result (Abstraction)
public interface BillingCallback {
void onBillingComplete(boolean success, String message, Bill bill);
}
public BillingThread(BillingService billingService, Bill bill, BillingCallback callback) {
super("BillingThread-" + bill.getBillId());
this.billingService = billingService;
this.bill = bill;
this.callback = callback;
this.success = false;
this.message = "";
}
@Override
public void run() {
try {
// Simulate processing time
System.out.println("[" + getName() + "] Processing bill " + bill.getBillId() + "...");
Thread.sleep(1000); // Simulate processing delay
// Process the bill
synchronized (billingService) {
success = billingService.createBill(bill);
}
if (success) {
message = "Bill " + bill.getBillId() + " processed successfully!";
System.out.println("[" + getName() + "] " + message);
} else {
message = "Failed to process bill " + bill.getBillId() + ". Insufficient stock.";
System.out.println("[" + getName() + "] " + message);
}
// Simulate invoice generation time
Thread.sleep(500);
} catch (InterruptedException e) {
message = "Billing interrupted for " + bill.getBillId();
Thread.currentThread().interrupt();
}
// Callback
if (callback != null) {
callback.onBillingComplete(success, message, bill);
}
}
public boolean isSuccess() {
return success;
}
public String getMessage() {
return message;
}
}
```

20. BackupService.java

Implements Runnable to perform periodic data backups. Calls DataStore.backupToText() for all five service classes to write human-readable backup files.

Code for BackupService.java:

```
package com.pharmacy.service;
/**
* Backup service that creates file backups of all data.
* Demonstrates: Runnable, Thread, FileWriter (via DataStore).
*/
public class BackupService implements Runnable {
private InventoryManager inventoryManager;
private CustomerService customerService;
private SupplierService supplierService;
private BillingService billingService;
private PrescriptionService prescriptionService;
private BackupCallback callback;
public interface BackupCallback {
void onBackupComplete(String message);
}
public BackupService(InventoryManager inventoryManager, CustomerService customerService,
SupplierService supplierService, BillingService billingService,
PrescriptionService prescriptionService) {
this.inventoryManager = inventoryManager;
this.customerService = customerService;
this.supplierService = supplierService;
this.billingService = billingService;
this.prescriptionService = prescriptionService;
}
public void setCallback(BackupCallback callback) {
this.callback = callback;
}
@Override
public void run() {
try {
System.out.println("[BackupService] Starting backup...");
Thread.sleep(500);
inventoryManager.createBackup();
Thread.sleep(200);
customerService.createBackup();
Thread.sleep(200);
supplierService.createBackup();
Thread.sleep(200);
billingService.createBackup();
Thread.sleep(200);
prescriptionService.createBackup();
Thread.sleep(200);
String msg = "All data backed up successfully!";
System.out.println("[BackupService] " + msg);
if (callback != null) {
callback.onBackupComplete(msg);
}
} catch (InterruptedException e) {
Thread.currentThread().interrupt();
```

```
if (callback != null) {
callback.onBackupComplete("Backup interrupted!");
}
}
}
}
```

UI Package (com.pharmacy.ui)

21. PharmacyApp.java

Main JavaFX Application entry point. Extends Application and overrides start(). Initializes all service objects, starts background threads, loads sample data on first run, and manages screen navigation using Stage.setScene().

Code for PharmacyApp.java:

```
package com.pharmacy.ui;

import com.pharmacy.model.User;
import com.pharmacy.service.*;
import com.pharmacy.util.SampleDataLoader;

import javafx.application.Application;
import javafx.application.Platform;
import javafx.stage.Stage;

import java.io.File;

/**
* Main JavaFX Application class - Entry point for the Pharmacy Management System.
* Demonstrates: JavaFX Application, Stage management, Inheritance (extends Application).
*/
public class PharmacyApp extends Application {

// Service instances (shared across all screens)
private UserService userService;
private InventoryManager inventoryManager;
private CustomerService customerService;
private SupplierService supplierService;
private BillingService billingService;
private PrescriptionService prescriptionService;

// Alert threads
private ExpiryAlertThread expiryAlertThread;
private Thread stockAlertThreadHandle;
private StockAlertThread stockAlertRunnable;

private Stage primaryStage;

@Override
public void start(Stage primaryStage) {
this.primaryStage = primaryStage;

// Create data directory
new File("data/invoices").mkdirs();

// Initialize all services
userService = new UserService();
inventoryManager = new InventoryManager();
customerService = new CustomerService();
supplierService = new SupplierService();
billingService = new BillingService(inventoryManager);
prescriptionService = new PrescriptionService();

// Load sample data if empty
SampleDataLoader.loadSampleData(inventoryManager, customerService, supplierService);
```

```
// Start background alert threads (Multithreading demonstration)
startAlertThreads();
// Show login screen
showLoginScreen();
primaryStage.setTitle("PharmaCare - Pharmacy Management System");
primaryStage.setWidth(1100);
primaryStage.setHeight(700);
primaryStage.setOnCloseRequest(e -> {
stopAlertThreads();
Platform.exit();
System.exit(0);
});
primaryStage.show();
}
/**
* Start background threads for monitoring.
* Demonstrates: Thread creation (both Thread subclass and Runnable), start(), daemon threads.
*/
private void startAlertThreads() {
// Thread extending Thread class
expiryAlertThread = new ExpiryAlertThread(inventoryManager);
expiryAlertThread.start();
// Thread using Runnable interface
stockAlertRunnable = new StockAlertThread(inventoryManager);
stockAlertThreadHandle = new Thread(stockAlertRunnable, "StockAlertThread");
stockAlertThreadHandle.setDaemon(true);
stockAlertThreadHandle.start();
}
private void stopAlertThreads() {
if (expiryAlertThread != null) {
expiryAlertThread.stopChecking();
}
if (stockAlertRunnable != null) {
stockAlertRunnable.stopChecking();
}
}
// Show login screen
public void showLoginScreen() {
LoginScreen loginScreen = new LoginScreen(this, userService);
primaryStage.setScene(loginScreen.getScene());
}
// Show dashboard after login
public void showDashboard(User user) {
DashboardScreen dashboard = new DashboardScreen(this, user);
primaryStage.setScene(dashboard.getScene());
}
// Show inventory module
public void showInventoryScreen() {
InventoryScreen screen = new InventoryScreen(this, inventoryManager, supplierService);
primaryStage.setScene(screen.getScene());
}
// Show billing module
public void showBillingScreen() {
BillingScreen screen = new BillingScreen(this, billingService, inventoryManager,
customerService, userService);
```

```
primaryStage.setScene(screen.getScene());
}
// Show supplier module
public void showSupplierScreen() {
SupplierScreen screen = new SupplierScreen(this, supplierService);
primaryStage.setScene(screen.getScene());
}
// Show customer module
public void showCustomerScreen() {
CustomerScreen screen = new CustomerScreen(this, customerService);
primaryStage.setScene(screen.getScene());
}
// Show prescription module
public void showPrescriptionScreen() {
PrescriptionScreen screen = new PrescriptionScreen(this, prescriptionService,
customerService, inventoryManager);
primaryStage.setScene(screen.getScene());
}
// Show reports module
public void showReportsScreen() {
ReportsScreen screen = new ReportsScreen(this, inventoryManager, billingService,
customerService, supplierService,
expiryAlertThread, stockAlertRunnable);
primaryStage.setScene(screen.getScene());
}
// Getters for services
public UserService getUserService() { return userService; }
public InventoryManager getInventoryManager() { return inventoryManager; }
public CustomerService getCustomerService() { return customerService; }
public SupplierService getSupplierService() { return supplierService; }
public BillingService getBillingService() { return billingService; }
public PrescriptionService getPrescriptionService() { return prescriptionService; }
public ExpiryAlertThread getExpiryAlertThread() { return expiryAlertThread; }
public StockAlertThread getStockAlertRunnable() { return stockAlertRunnable; }
public Stage getPrimaryStage() { return primaryStage; }
/**
* Main method - entry point.
*/
public static void main(String[] args) {
launch(args);
}
}
```

22. LoginScreen.java

Login interface built with GridPane. Uses PasswordField to mask passwords and validates credentials via UserService. Supports Admin and Pharmacist roles and redirects to the Dashboard on successful login.

Code for LoginScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.User;
import com.pharmacy.service.UserService;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
```

```
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Login Screen with role-based authentication.
* Demonstrates: JavaFX (GridPane, TextField, Button, Label), Event handling.
*/
public class LoginScreen {
private Scene scene;
private PharmacyApp app;
private UserService userService;
public LoginScreen(PharmacyApp app, UserService userService) {
this.app = app;
this.userService = userService;
createScene();
}
private void createScene() {
// Main layout
VBox mainLayout = new VBox(20);
mainLayout.setAlignment(Pos.CENTER);
mainLayout.setPadding(new Insets(50));
mainLayout.setStyle("-fx-background-color: #2c3e50;");
// Title
Label titleLabel = new Label("PharmaCare");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 36));
titleLabel.setStyle("-fx-text-fill: #ecf0f1;");
Label subtitleLabel = new Label("Pharmacy Management System");
subtitleLabel.setFont(Font.font("Arial", FontWeight.NORMAL, 16));
subtitleLabel.setStyle("-fx-text-fill: #bdc3c7;");
// Login form using GridPane
GridPane grid = new GridPane();
grid.setAlignment(Pos.CENTER);
grid.setHgap(10);
grid.setVgap(15);
grid.setPadding(new Insets(30));
grid.setMaxWidth(400);
grid.setStyle("-fx-background-color: #34495e; -fx-background-radius: 10;");
Label loginTitle = new Label("Login");
loginTitle.setFont(Font.font("Arial", FontWeight.BOLD, 20));
loginTitle.setStyle("-fx-text-fill: #ecf0f1;");
grid.add(loginTitle, 0, 0, 2, 1);
Label userLabel = new Label("Username:");
userLabel.setStyle("-fx-text-fill: #ecf0f1;");
TextField usernameField = new TextField();
usernameField.setPromptText("Enter username");
usernameField.setPrefWidth(250);
grid.add(userLabel, 0, 1);
grid.add(usernameField, 1, 1);
Label passLabel = new Label("Password:");
passLabel.setStyle("-fx-text-fill: #ecf0f1;");
PasswordField passwordField = new PasswordField();
passwordField.setPromptText("Enter password");
grid.add(passLabel, 0, 2);
grid.add(passwordField, 1, 2);
```

```
Label messageLabel = new Label();
messageLabel.setStyle("-fx-text-fill: #e74c3c;");
Button loginBtn = new Button("Login");
loginBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; " +
"-fx-font-size: 14; -fx-pref-width: 250; -fx-cursor: hand;");
loginBtn.setOnAction(e -> {
String username = usernameField.getText().trim();
String password = passwordField.getText().trim();
if (username.isEmpty() || password.isEmpty()) {
messageLabel.setText("Please enter both username and password.");
return;
}
User user = userService.authenticate(username, password);
if (user != null) {
app.showDashboard(user);
} else {
messageLabel.setText("Invalid credentials! Please try again.");
passwordField.clear();
}
});
// Allow Enter key to login
passwordField.setOnAction(e -> loginBtn.fire());
usernameField.setOnAction(e -> loginBtn.fire());
grid.add(loginBtn, 0, 3, 2, 1);
grid.add(messageLabel, 0, 4, 2, 1);
// Default credentials hint
Label hintLabel = new Label("Default: admin/admin123 or pharmacist/pharma123");
hintLabel.setStyle("-fx-text-fill: #7f8c8d; -fx-font-size: 11;");
mainLayout.getChildren().addAll(titleLabel, subtitleLabel, grid, hintLabel);
scene = new Scene(mainLayout, 1100, 700);
}
public Scene getScene() {
return scene;
}
}
```

23. DashboardScreen.java

Main dashboard displayed after login. Uses BorderPane with an HBox top bar and a GridPane of six navigation buttons. Displays the logged-in user's name and role, and provides a Logout button.

Code for DashboardScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.User;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Dashboard screen - main navigation hub after login.
* Demonstrates: JavaFX (VBox, HBox, Button, Label, GridPane), Event handling.
*/
```

```
public class DashboardScreen {
private Scene scene;
private PharmacyApp app;
private User currentUser;
public DashboardScreen(PharmacyApp app, User currentUser) {
this.app = app;
this.currentUser = currentUser;
createScene();
}
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(20);
topBar.setPadding(new Insets(15, 25, 15, 25));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Label titleLabel = new Label("PharmaCare Dashboard");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 22));
titleLabel.setStyle("-fx-text-fill: white;");
Region spacer = new Region();
HBox.setHgrow(spacer, Priority.ALWAYS);
Label userLabel = new Label("Welcome, " + currentUser.getFullName()
+ " (" + currentUser.getRole().getDisplayName() + ")");
userLabel.setStyle("-fx-text-fill: #bdc3c7; -fx-font-size: 13;");
Button logoutBtn = new Button("Logout");
logoutBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
logoutBtn.setOnAction(e -> {
app.getUserService().logout();
app.showLoginScreen();
});
topBar.getChildren().addAll(titleLabel, spacer, userLabel, logoutBtn);
root.setTop(topBar);
// Center - Module buttons in a grid
GridPane moduleGrid = new GridPane();
moduleGrid.setAlignment(Pos.CENTER);
moduleGrid.setHgap(25);
moduleGrid.setVgap(25);
moduleGrid.setPadding(new Insets(40));
// Module buttons
Button inventoryBtn = createModuleButton("Inventory\nManagement",
"Manage medicines, stock,\nbatch & expiry tracking", "#3498db");
inventoryBtn.setOnAction(e -> app.showInventoryScreen());
Button billingBtn = createModuleButton("Sales &\nBilling",
"Create bills, generate\ninvoices for customers", "#27ae60");
billingBtn.setOnAction(e -> app.showBillingScreen());
Button supplierBtn = createModuleButton("Supplier\nManagement",
"Manage supplier records\nand contact details", "#e67e22");
supplierBtn.setOnAction(e -> app.showSupplierScreen());
Button customerBtn = createModuleButton("Customer\nRecords",
"Manage patient/customer\ninformation", "#9b59b6");
customerBtn.setOnAction(e -> app.showCustomerScreen());
Button prescriptionBtn = createModuleButton("Prescription\nHandling",
"Record and manage\nprescription details", "#1abc9c");
```

```
prescriptionBtn.setOnAction(e -> app.showPrescriptionScreen());
Button reportsBtn = createModuleButton("Reports &\nAlerts",
"View sales reports, stock\n& expiry alerts", "#e74c3c");
reportsBtn.setOnAction(e -> app.showReportsScreen());
// Arrange in 3x2 grid
moduleGrid.add(inventoryBtn, 0, 0);
moduleGrid.add(billingBtn, 1, 0);
moduleGrid.add(supplierBtn, 2, 0);
moduleGrid.add(customerBtn, 0, 1);
moduleGrid.add(prescriptionBtn, 1, 1);
moduleGrid.add(reportsBtn, 2, 1);
root.setCenter(moduleGrid);
// Bottom status bar
HBox statusBar = new HBox(20);
statusBar.setPadding(new Insets(10, 25, 10, 25));
statusBar.setStyle("-fx-background-color: #34495e;");
Label statusLabel = new Label("Medicines: " + app.getInventoryManager().getMedicineCount()
+ " | Customers: " + app.getCustomerService().getCustomerCount()
+ " | Suppliers: " + app.getSupplierService().getSupplierCount()
+ " | Bills: " + app.getBillingService().getBillCount());
statusLabel.setStyle("-fx-text-fill: #bdc3c7; -fx-font-size: 12;");
statusBar.getChildren().add(statusLabel);
root.setBottom(statusBar);
scene = new Scene(root, 1100, 700);
}
// Helper to create styled module buttons
private Button createModuleButton(String title, String description, String color) {
VBox content = new VBox(8);
content.setAlignment(Pos.CENTER);
Label titleLbl = new Label(title);
titleLbl.setFont(Font.font("Arial", FontWeight.BOLD, 16));
titleLbl.setStyle("-fx-text-fill: white; -fx-text-alignment: center;");
Label descLbl = new Label(description);
descLbl.setStyle("-fx-text-fill: #ecf0f1; -fx-font-size: 11; -fx-text-alignment: center;");
content.getChildren().addAll(titleLbl, descLbl);
Button btn = new Button();
btn.setGraphic(content);
btn.setPrefSize(230, 130);
btn.setStyle("-fx-background-color: " + color + "; -fx-background-radius: 10; -fx-cursor:
hand;");
return btn;
}
public Scene getScene() {
return scene;
}
}
```

24. InventoryScreen.java

Full inventory management screen with a TableView showing all medicines. Provides Add, Edit, Delete, Search by name, and Filter by category using a ComboBox. Supports sorting by clicking column headers.

Code for InventoryScreen.java:

```
package com.pharmacy.ui;
```

```
import com.pharmacy.model.Medicine;
import com.pharmacy.model.MedicineCategory;
import com.pharmacy.service.InventoryManager;
import com.pharmacy.service.SupplierService;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.util.ArrayList;
/**
* Inventory Management Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Buttons, ComboBox.
*/
public class InventoryScreen {
private Scene scene;
private PharmacyApp app;
private InventoryManager inventoryManager;
private SupplierService supplierService;
private TableView<Medicine> tableView;
private ObservableList<Medicine> tableData;
public InventoryScreen(PharmacyApp app, InventoryManager inventoryManager,
SupplierService supplierService) {
this.app = app;
this.inventoryManager = inventoryManager;
this.supplierService = supplierService;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = createTopBar("Inventory Management");
root.setTop(topBar);
// Left side - Form
VBox formBox = createForm();
root.setLeft(formBox);
// Center - Table
VBox tableBox = createTable();
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private HBox createTopBar(String title) {
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
```

```
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));

Label titleLabel = new Label(title);
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");

topBar.getChildren().addAll(backBtn, titleLabel);
return topBar;
}

private VBox createForm() {
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");

Label formTitle = new Label("Add / Edit Medicine");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));

// Form fields using GridPane
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);

TextField idField = new TextField();
idField.setEditable(false);
idField.setPromptText("Auto-generated");

TextField nameField = new TextField();
nameField.setPromptText("Medicine name");

TextField mfgField = new TextField();
mfgField.setPromptText("Manufacturer");

ComboBox<MedicineCategory> categoryBox = new ComboBox<>();
categoryBox.getItems().addAll(MedicineCategory.values());
categoryBox.setPromptText("Select Category");
categoryBox.setPrefWidth(180);

TextField batchField = new TextField();
batchField.setPromptText("Batch number");

TextField expiryField = new TextField();
expiryField.setPromptText("YYYY-MM-DD");

TextField qtyField = new TextField();
qtyField.setPromptText("Quantity");

TextField priceField = new TextField();
priceField.setPromptText("Price (Rs.)");

TextField supplierField = new TextField();
supplierField.setPromptText("Supplier ID");

int row = 0;
grid.add(new Label("ID:"), 0, row); grid.add(idField, 1, row++);
grid.add(new Label("Name:"), 0, row); grid.add(nameField, 1, row++);
grid.add(new Label("Manufacturer:"), 0, row); grid.add(mfgField, 1, row++);
grid.add(new Label("Category:"), 0, row); grid.add(categoryBox, 1, row++);
grid.add(new Label("Batch No:"), 0, row); grid.add(batchField, 1, row++);
grid.add(new Label("Expiry Date:"), 0, row); grid.add(expiryField, 1, row++);
grid.add(new Label("Quantity:"), 0, row); grid.add(qtyField, 1, row++);
grid.add(new Label("Price:"), 0, row); grid.add(priceField, 1, row++);
grid.add(new Label("Supplier ID:"), 0, row); grid.add(supplierField, 1, row++);

Label statusLabel = new Label();
statusLabel.setStyle("-fx-text-fill: #27ae60;");

// Buttons
```

```
HBox btnBox = new HBox(8);
Button addBtn = new Button("Add");
addBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-cursor: hand;");
Button updateBtn = new Button("Update");
updateBtn.setStyle("-fx-background-color: #2980b9; -fx-text-fill: white; -fx-cursor: hand;");
Button deleteBtn = new Button("Delete");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
Button clearBtn = new Button("Clear");
clearBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
btnBox.getChildren().addAll(addBtn, updateBtn, deleteBtn, clearBtn);
// Add button action
addBtn.setOnAction(e -> {
try {
String id = IDGenerator.generateMedicineId();
Medicine med = new Medicine(id, nameField.getText(), mfgField.getText(),
categoryBox.getValue(), batchField.getText(), expiryField.getText(),
Integer.parseInt(qtyField.getText()), // Wrapper class parsing
Double.parseDouble(priceField.getText()), // Wrapper class parsing
supplierField.getText());
inventoryManager.addMedicine(med);
refreshTable();
clearFields(idField, nameField, mfgField, categoryBox, batchField,
expiryField, qtyField, priceField, supplierField);
statusLabel.setText("Medicine added: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
} catch (Exception ex) {
statusLabel.setText("Error: " + ex.getMessage());
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
});
// Update button action
updateBtn.setOnAction(e -> {
try {
String id = idField.getText();
if (id.isEmpty()) {
statusLabel.setText("Select a medicine first!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
Medicine med = new Medicine(id, nameField.getText(), mfgField.getText(),
categoryBox.getValue(), batchField.getText(), expiryField.getText(),
Integer.parseInt(qtyField.getText()),
Double.parseDouble(priceField.getText()),
supplierField.getText());
inventoryManager.updateMedicine(med);
refreshTable();
statusLabel.setText("Medicine updated: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
} catch (Exception ex) {
statusLabel.setText("Error: " + ex.getMessage());
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
});
// Delete button action
deleteBtn.setOnAction(e -> {
String id = idField.getText();
```

```
if (!id.isEmpty()) {
inventoryManager.deleteMedicine(id);
refreshTable();
clearFields(idField, nameField, mfgField, categoryBox, batchField,
expiryField, qtyField, priceField, supplierField);
statusLabel.setText("Medicine deleted: " + id);
}
});
// Clear button
clearBtn.setOnAction(e -> {
clearFields(idField, nameField, mfgField, categoryBox, batchField,
expiryField, qtyField, priceField, supplierField);
statusLabel.setText("");
});
// Table selection listener - populate form
tableView = new TableView<>();
tableView.getSelectionModel().selectedItemProperty().addListener((obs, oldVal, newVal) -> {
if (newVal != null) {
idField.setText(newVal.getMedicineId());
nameField.setText(newVal.getName());
mfgField.setText(newVal.getManufacturer());
categoryBox.setValue(newVal.getCategory());
batchField.setText(newVal.getBatchNumber());
expiryField.setText(newVal.getExpiryDate());
qtyField.setText(String.valueOf(newVal.getQuantity()));
priceField.setText(String.valueOf(newVal.getPrice()));
supplierField.setText(newVal.getSupplierId());
}
});
formBox.getChildren().addAll(formTitle, grid, btnBox, statusLabel);
return formBox;
}
@SuppressWarnings("unchecked")
private VBox createTable() {
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
// Search and sort controls
HBox controlsBox = new HBox(10);
controlsBox.setAlignment(Pos.CENTER_LEFT);
TextField searchField = new TextField();
searchField.setPromptText("Search by name...");
searchField.setPrefWidth(200);
Button searchBtn = new Button("Search");
searchBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;");
searchBtn.setOnAction(e -> {
String query = searchField.getText().trim();
if (query.isEmpty()) {
refreshTable();
} else {
ArrayList<Medicine> results = inventoryManager.searchByName(query);
tableData = FXCollections.observableArrayList(results);
tableView.setItems(tableData);
}
});
ComboBox<String> sortBox = new ComboBox<>();
sortBox.getItems().addAll("Sort by Name", "Sort by Price", "Sort by Expiry");
```

```
sortBox.setPromptText("Sort by...");
sortBox.setOnAction(e -> {
String selected = sortBox.getValue();
ArrayList<Medicine> sorted;
if ("Sort by Name".equals(selected)) {
sorted = inventoryManager.getSortedByName();
} else if ("Sort by Price".equals(selected)) {
sorted = inventoryManager.getSortedByPrice();
} else {
sorted = inventoryManager.getSortedByExpiry();
}
tableData = FXCollections.observableArrayList(sorted);
tableView.setItems(tableData);
});
ComboBox<MedicineCategory> filterBox = new ComboBox<>();
filterBox.getItems().addAll(MedicineCategory.values());
filterBox.setPromptText("Filter by Category");
filterBox.setOnAction(e -> {
MedicineCategory cat = filterBox.getValue();
if (cat != null) {
ArrayList<Medicine> filtered = inventoryManager.filterByCategory(cat);
tableData = FXCollections.observableArrayList(filtered);
tableView.setItems(tableData);
}
});
Button refreshBtn = new Button("Show All");
refreshBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
refreshBtn.setOnAction(e -> refreshTable());
controlsBox.getChildren().addAll(searchField, searchBtn, sortBox, filterBox, refreshBtn);
// Table columns
TableColumn<Medicine, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("medicineId"));
idCol.setPrefWidth(80);
TableColumn<Medicine, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(150);
TableColumn<Medicine, MedicineCategory> catCol = new TableColumn<>("Category");
catCol.setCellValueFactory(new PropertyValueFactory<>("category"));
catCol.setPrefWidth(80);
TableColumn<Medicine, String> batchCol = new TableColumn<>("Batch");
batchCol.setCellValueFactory(new PropertyValueFactory<>("batchNumber"));
batchCol.setPrefWidth(70);
TableColumn<Medicine, String> expiryCol = new TableColumn<>("Expiry");
expiryCol.setCellValueFactory(new PropertyValueFactory<>("expiryDate"));
expiryCol.setPrefWidth(90);
TableColumn<Medicine, Integer> qtyCol = new TableColumn<>("Qty");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
qtyCol.setPrefWidth(50);
TableColumn<Medicine, Double> priceCol = new TableColumn<>("Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("price"));
priceCol.setPrefWidth(70);
tableView.getColumns().addAll(idCol, nameCol, catCol, batchCol, expiryCol, qtyCol, priceCol);
refreshTable();
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(controlsBox, tableView);
```

```
return tableBox;
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(inventoryManager.getAllMedicines());
tableView.setItems(tableData);
}
private void clearFields(TextField idField, TextField nameField, TextField mfgField,
ComboBox<MedicineCategory> categoryBox, TextField batchField,
TextField expiryField, TextField qtyField, TextField priceField,
TextField supplierField) {
idField.clear();
nameField.clear();
mfgField.clear();
categoryBox.setValue(null);
batchField.clear();
expiryField.clear();
qtyField.clear();
priceField.clear();
supplierField.clear();
tableView.getSelectionModel().clearSelection();
}
public Scene getScene() {
return scene;
}
}
```

25. BillingScreen.java

Billing and sales screen. Allows staff to select a customer, add medicine items to a bill, apply discounts, and process payment. Uses BillingThread for async processing and Platform.runLater() to safely update the UI.

Code for BillingScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.*;
import com.pharmacy.service.*;
import com.pharmacy.util.IDGenerator;
import javafx.application.Platform;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.text.SimpleDateFormat;
import java.util.Date;
/**
* Sales and Billing Screen with multi-threaded billing simulation.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Multithreaded billing.
*/
public class BillingScreen {
private Scene scene;
```

```
private PharmacyApp app;
private BillingService billingService;
private InventoryManager inventoryManager;
private CustomerService customerService;
private UserService userService;
private TableView<BillItem> billItemsTable;
private ObservableList<BillItem> billItems;
private Label totalLabel;
private Label netLabel;
private Bill currentBill;
public BillingScreen(PharmacyApp app, BillingService billingService,
InventoryManager inventoryManager, CustomerService customerService,
UserService userService) {
this.app = app;
this.billingService = billingService;
this.inventoryManager = inventoryManager;
this.customerService = customerService;
this.userService = userService;
this.billItems = FXCollections.observableArrayList();
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(userService.getCurrentUser()));
Label titleLabel = new Label("Sales & Billing");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Bill form
VBox formBox = new VBox(12);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("New Bill");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane custGrid = new GridPane();
custGrid.setVgap(8);
custGrid.setHgap(8);
TextField custIdField = new TextField();
custIdField.setPromptText("Customer ID");
TextField custNameField = new TextField();
custNameField.setPromptText("Customer Name");
// Auto-fill customer name when ID is entered
custIdField.textProperty().addListener((obs, oldVal, newVal) -> {
```

```
if (newVal != null && !newVal.isEmpty()) {
Customer cust = customerService.getCustomerById(newVal.trim());
if (cust != null) {
custNameField.setText(cust.getName());
}
}
});
custGrid.add(new Label("Customer ID:"), 0, 0);
custGrid.add(custIdField, 1, 0);
custGrid.add(new Label("Customer Name:"), 0, 1);
custGrid.add(custNameField, 1, 1);
// Add medicine to bill
Label addItemTitle = new Label("Add Medicine to Bill");
addItemTitle.setFont(Font.font("Arial", FontWeight.BOLD, 13));
GridPane itemGrid = new GridPane();
itemGrid.setVgap(8);
itemGrid.setHgap(8);
TextField medIdField = new TextField();
medIdField.setPromptText("Medicine ID");
Label medNameLabel = new Label("-");
Label medPriceLabel = new Label("-");
Label medStockLabel = new Label("-");
TextField medQtyField = new TextField();
medQtyField.setPromptText("Quantity");
// Auto-fill medicine details
medIdField.textProperty().addListener((obs, oldVal, newVal) -> {
if (newVal != null && !newVal.isEmpty()) {
Medicine med = inventoryManager.getMedicineById(newVal.trim());
if (med != null) {
medNameLabel.setText(med.getName());
medPriceLabel.setText("Rs." + String.format("%.2f", med.getPrice()));
medStockLabel.setText("Stock: " + med.getQuantity());
}
}
});
int r = 0;
itemGrid.add(new Label("Medicine ID:"), 0, r); itemGrid.add(medIdField, 1, r++);
itemGrid.add(new Label("Name:"), 0, r); itemGrid.add(medNameLabel, 1, r++);
itemGrid.add(new Label("Price:"), 0, r); itemGrid.add(medPriceLabel, 1, r++);
itemGrid.add(new Label("Available:"), 0, r); itemGrid.add(medStockLabel, 1, r++);
itemGrid.add(new Label("Quantity:"), 0, r); itemGrid.add(medQtyField, 1, r++);
Label statusLabel = new Label();
Button addItemBtn = new Button("Add to Bill");
addItemBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;
-fx-pref-width: 200;");
addItemBtn.setOnAction(e -> {
try {
String medId = medIdField.getText().trim();
Medicine med = inventoryManager.getMedicineById(medId);
if (med == null) {
statusLabel.setText("Medicine not found!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
int qty = Integer.parseInt(medQtyField.getText().trim());
if (qty > med.getQuantity()) {
```

```
statusLabel.setText("Insufficient stock!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
BillItem item = new BillItem(medId, med.getName(), qty, med.getPrice());
billItems.add(item);
updateTotals();
medIdField.clear();
medQtyField.clear();
medNameLabel.setText("-");
medPriceLabel.setText("-");
medStockLabel.setText("-");
statusLabel.setText("Item added!");
statusLabel.setStyle("-fx-text-fill: #27ae60;");
} catch (NumberFormatException ex) {
statusLabel.setText("Invalid quantity!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
});
// Totals
totalLabel = new Label("Total: Rs.0.00");
totalLabel.setFont(Font.font("Arial", FontWeight.BOLD, 14));
TextField discountField = new TextField("0");
discountField.setPromptText("Discount %");
discountField.setPrefWidth(80);
netLabel = new Label("Net: Rs.0.00");
netLabel.setFont(Font.font("Arial", FontWeight.BOLD, 16));
netLabel.setStyle("-fx-text-fill: #27ae60;");
HBox totalsBox = new HBox(10);
totalsBox.setAlignment(Pos.CENTER_LEFT);
totalsBox.getChildren().addAll(new Label("Discount %:"), discountField);
// Generate Bill button - uses multithreaded billing
Button generateBtn = new Button("Generate Bill (Threaded)");
generateBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; " +
"-fx-font-size: 14; -fx-pref-width: 280; -fx-cursor: hand;");
generateBtn.setOnAction(e -> {
if (billItems.isEmpty()) {
statusLabel.setText("Add items first!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
if (custNameField.getText().trim().isEmpty()) {
statusLabel.setText("Enter customer name!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
String billId = IDGenerator.generateBillId();
String date = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
currentBill = new Bill(billId, custIdField.getText().trim(),
custNameField.getText().trim(), date,
userService.getCurrentUser().getUsername());
for (BillItem item : billItems) {
currentBill.addItem(item);
}
try {
```

```
double discPct = Double.parseDouble(discountField.getText().trim());
currentBill.applyDiscount(discPct); // Method overloading demo
} catch (NumberFormatException ex) {
// No discount
}
// Multi-threaded billing using BillingThread
statusLabel.setText("Processing bill...");
statusLabel.setStyle("-fx-text-fill: #3498db;");
generateBtn.setDisable(true);
BillingThread billingThread = new BillingThread(billingService, currentBill,
(success, message, bill) -> {
// Use Platform.runLater to update UI from thread
Platform.runLater(() -> {
if (success) {
statusLabel.setText(message + " Invoice generated!");
statusLabel.setStyle("-fx-text-fill: #27ae60;");
billItems.clear();
updateTotals();
custIdField.clear();
custNameField.clear();
} else {
statusLabel.setText(message);
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
generateBtn.setDisable(false);
});
});
billingThread.start();
// Demonstrate join() - wait for billing thread in another thread
Thread waitThread = new Thread(() -> {
try {
billingThread.join(); // Wait for billing to complete
System.out.println("Billing thread completed via join()");
} catch (InterruptedException ex) {
Thread.currentThread().interrupt();
}
});
waitThread.setDaemon(true);
waitThread.start();
});
Button clearBillBtn = new Button("Clear Bill");
clearBillBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor:
hand;");
clearBillBtn.setOnAction(e -> {
billItems.clear();
updateTotals();
statusLabel.setText("");
});
formBox.getChildren().addAll(formTitle, custGrid,
new Separator(), addItemTitle, itemGrid, addItemBtn, statusLabel,
new Separator(), totalLabel, totalsBox, netLabel,
generateBtn, clearBillBtn);
root.setLeft(formBox);
// Center - Bill items table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
```

```
Label tableTitle = new Label("Current Bill Items");
tableTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
billItemsTable = new TableView<>();
billItemsTable.setItems(billItems);
TableColumn<BillItem, String> nameCol = new TableColumn<>("Medicine");
nameCol.setCellValueFactory(new PropertyValueFactory<>("medicineName"));
nameCol.setPrefWidth(200);
TableColumn<BillItem, Integer> qtyCol = new TableColumn<>("Qty");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
qtyCol.setPrefWidth(60);
TableColumn<BillItem, Double> priceCol = new TableColumn<>("Unit Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("unitPrice"));
priceCol.setPrefWidth(100);
TableColumn<BillItem, Double> subCol = new TableColumn<>("Subtotal");
subCol.setCellValueFactory(new PropertyValueFactory<>("subtotal"));
subCol.setPrefWidth(100);
billItemsTable.getColumns().addAll(nameCol, qtyCol, priceCol, subCol);
// Past bills table
Label pastTitle = new Label("Past Bills");
pastTitle.setFont(Font.font("Arial", FontWeight.BOLD, 13));
TableView<Bill> pastBillsTable = new TableView<>();
ObservableList<Bill> pastBills =
FXCollections.observableArrayList(billingService.getAllBills());
pastBillsTable.setItems(pastBills);
TableColumn<Bill, String> billIdCol = new TableColumn<>("Bill ID");
billIdCol.setCellValueFactory(new PropertyValueFactory<>("billId"));
TableColumn<Bill, String> billCustCol = new TableColumn<>("Customer");
billCustCol.setCellValueFactory(new PropertyValueFactory<>("customerName"));
TableColumn<Bill, String> billDateCol = new TableColumn<>("Date");
billDateCol.setCellValueFactory(new PropertyValueFactory<>("date"));
TableColumn<Bill, Double> billAmtCol = new TableColumn<>("Amount");
billAmtCol.setCellValueFactory(new PropertyValueFactory<>("netAmount"));
pastBillsTable.getColumns().addAll(billIdCol, billCustCol, billDateCol, billAmtCol);
pastBillsTable.setPrefHeight(200);
VBox.setVgrow(billItemsTable, Priority.ALWAYS);
tableBox.getChildren().addAll(tableTitle, billItemsTable, pastTitle, pastBillsTable);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void updateTotals() {
double total = 0;
for (BillItem item : billItems) {
total += item.getSubtotal();
}
totalLabel.setText(String.format("Total: Rs.%.2f", total));
netLabel.setText(String.format("Net: Rs.%.2f", total));
}
public Scene getScene() {
return scene;
}
}
```

26. SupplierScreen.java

Supplier management screen with a TableView and full CRUD operations. Allows adding, editing, and deleting supplier records with input validation.

Code for SupplierScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Supplier;
import com.pharmacy.service.SupplierService;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Supplier Management Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Event handling.
*/
public class SupplierScreen {
private Scene scene;
private PharmacyApp app;
private SupplierService supplierService;
private TableView<Supplier> tableView;
private ObservableList<Supplier> tableData;
public SupplierScreen(PharmacyApp app, SupplierService supplierService) {
this.app = app;
this.supplierService = supplierService;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Supplier Management");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Form
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
```

```
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("Add / Edit Supplier");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField idField = new TextField();
idField.setEditable(false);
idField.setPromptText("Auto-generated");
TextField companyField = new TextField();
companyField.setPromptText("Company name");
TextField contactField = new TextField();
contactField.setPromptText("Contact person");
TextField phoneField = new TextField();
phoneField.setPromptText("Phone");
TextField emailField = new TextField();
emailField.setPromptText("Email");
TextField addressField = new TextField();
addressField.setPromptText("Address");
int row = 0;
grid.add(new Label("ID:"), 0, row); grid.add(idField, 1, row++);
grid.add(new Label("Company:"), 0, row); grid.add(companyField, 1, row++);
grid.add(new Label("Contact:"), 0, row); grid.add(contactField, 1, row++);
grid.add(new Label("Phone:"), 0, row); grid.add(phoneField, 1, row++);
grid.add(new Label("Email:"), 0, row); grid.add(emailField, 1, row++);
grid.add(new Label("Address:"), 0, row); grid.add(addressField, 1, row++);
Label statusLabel = new Label();
HBox btnBox = new HBox(8);
Button addBtn = new Button("Add");
addBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-cursor: hand;");
Button updateBtn = new Button("Update");
updateBtn.setStyle("-fx-background-color: #2980b9; -fx-text-fill: white; -fx-cursor: hand;");
Button deleteBtn = new Button("Delete");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
Button clearBtn = new Button("Clear");
clearBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
btnBox.getChildren().addAll(addBtn, updateBtn, deleteBtn, clearBtn);
addBtn.setOnAction(e -> {
String id = IDGenerator.generateSupplierId();
Supplier sup = new Supplier(id, companyField.getText(), contactField.getText(),
phoneField.getText(), emailField.getText(), addressField.getText());
supplierService.addSupplier(sup);
refreshTable();
clearFields(idField, companyField, contactField, phoneField, emailField, addressField);
statusLabel.setText("Supplier added: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
updateBtn.setOnAction(e -> {
String id = idField.getText();
if (id.isEmpty()) { statusLabel.setText("Select a supplier!"); return; }
Supplier sup = new Supplier(id, companyField.getText(), contactField.getText(),
phoneField.getText(), emailField.getText(), addressField.getText());
supplierService.updateSupplier(sup);
refreshTable();
statusLabel.setText("Supplier updated: " + id);
```

```
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
deleteBtn.setOnAction(e -> {
String id = idField.getText();
if (!id.isEmpty()) {
supplierService.deleteSupplier(id);
refreshTable();
clearFields(idField, companyField, contactField, phoneField, emailField, addressField);
statusLabel.setText("Supplier deleted.");
}
});
clearBtn.setOnAction(e -> {
clearFields(idField, companyField, contactField, phoneField, emailField, addressField);
statusLabel.setText("");
});
formBox.getChildren().addAll(formTitle, grid, btnBox, statusLabel);
root.setLeft(formBox);
// Center - Table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
// Search
HBox searchBox = new HBox(10);
TextField searchField = new TextField();
searchField.setPromptText("Search by company name...");
Button searchBtn = new Button("Search");
searchBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;");
searchBtn.setOnAction(e -> {
String q = searchField.getText().trim();
if (q.isEmpty()) refreshTable();
else {
tableData = FXCollections.observableArrayList(supplierService.searchByName(q));
tableView.setItems(tableData);
}
});
Button showAllBtn = new Button("Show All");
showAllBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
showAllBtn.setOnAction(e -> refreshTable());
searchBox.getChildren().addAll(searchField, searchBtn, showAllBtn);
tableView = new TableView<>();
TableColumn<Supplier, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("supplierId"));
TableColumn<Supplier, String> compCol = new TableColumn<>("Company");
compCol.setCellValueFactory(new PropertyValueFactory<>("companyName"));
compCol.setPrefWidth(150);
TableColumn<Supplier, String> contCol = new TableColumn<>("Contact");
contCol.setCellValueFactory(new PropertyValueFactory<>("contactPerson"));
contCol.setPrefWidth(120);
TableColumn<Supplier, String> phoneCol = new TableColumn<>("Phone");
phoneCol.setCellValueFactory(new PropertyValueFactory<>("phone"));
phoneCol.setPrefWidth(110);
TableColumn<Supplier, String> emailCol = new TableColumn<>("Email");
emailCol.setCellValueFactory(new PropertyValueFactory<>("email"));
emailCol.setPrefWidth(150);
tableView.getColumns().addAll(idCol, compCol, contCol, phoneCol, emailCol);
refreshTable();
```

```
// Table selection listener
tableView.getSelectionModel().selectedItemProperty().addListener((obs, o, n) -> {
if (n != null) {
idField.setText(n.getSupplierId());
companyField.setText(n.getCompanyName());
contactField.setText(n.getContactPerson());
phoneField.setText(n.getPhone());
emailField.setText(n.getEmail());
addressField.setText(n.getAddress());
}
});
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(searchBox, tableView);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(supplierService.getAllSuppliers());
tableView.setItems(tableData);
}
private void clearFields(TextField... fields) {
for (TextField f : fields) f.clear();
tableView.getSelectionModel().clearSelection();
}
public Scene getScene() { return scene; }
}
```

27. CustomerScreen.java

Customer records screen with a TableView and full CRUD operations. Allows adding, editing, and deleting customer records with input validation.

Code for CustomerScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Customer;
import com.pharmacy.service.CustomerService;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Customer/Patient Records Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Event handling.
*/
public class CustomerScreen {
private Scene scene;
private PharmacyApp app;
private CustomerService customerService;
private TableView<Customer> tableView;
```

```
private ObservableList<Customer> tableData;
public CustomerScreen(PharmacyApp app, CustomerService customerService) {
this.app = app;
this.customerService = customerService;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Customer / Patient Records");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Form
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("Add / Edit Customer");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField idField = new TextField();
idField.setEditable(false);
idField.setPromptText("Auto-generated");
TextField nameField = new TextField();
nameField.setPromptText("Full name");
TextField phoneField = new TextField();
phoneField.setPromptText("Phone number");
TextField emailField = new TextField();
emailField.setPromptText("Email");
TextField addressField = new TextField();
addressField.setPromptText("Address");
int row = 0;
grid.add(new Label("ID:"), 0, row); grid.add(idField, 1, row++);
grid.add(new Label("Name:"), 0, row); grid.add(nameField, 1, row++);
grid.add(new Label("Phone:"), 0, row); grid.add(phoneField, 1, row++);
grid.add(new Label("Email:"), 0, row); grid.add(emailField, 1, row++);
grid.add(new Label("Address:"), 0, row); grid.add(addressField, 1, row++);
Label statusLabel = new Label();
HBox btnBox = new HBox(8);
Button addBtn = new Button("Add");
addBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-cursor: hand;");
```

```
Button updateBtn = new Button("Update");
updateBtn.setStyle("-fx-background-color: #2980b9; -fx-text-fill: white; -fx-cursor: hand;");
Button deleteBtn = new Button("Delete");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
Button clearBtn = new Button("Clear");
clearBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
btnBox.getChildren().addAll(addBtn, updateBtn, deleteBtn, clearBtn);
addBtn.setOnAction(e -> {
String id = IDGenerator.generateCustomerId();
Customer cust = new Customer(id, nameField.getText(), phoneField.getText(),
emailField.getText(), addressField.getText());
customerService.addCustomer(cust);
refreshTable();
clearFields(idField, nameField, phoneField, emailField, addressField);
statusLabel.setText("Customer added: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
updateBtn.setOnAction(e -> {
String id = idField.getText();
if (id.isEmpty()) { statusLabel.setText("Select a customer!"); return; }
Customer cust = new Customer(id, nameField.getText(), phoneField.getText(),
emailField.getText(), addressField.getText());
customerService.updateCustomer(cust);
refreshTable();
statusLabel.setText("Customer updated: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
deleteBtn.setOnAction(e -> {
String id = idField.getText();
if (!id.isEmpty()) {
customerService.deleteCustomer(id);
refreshTable();
clearFields(idField, nameField, phoneField, emailField, addressField);
statusLabel.setText("Customer deleted.");
}
});
clearBtn.setOnAction(e -> {
clearFields(idField, nameField, phoneField, emailField, addressField);
statusLabel.setText("");
});
formBox.getChildren().addAll(formTitle, grid, btnBox, statusLabel);
root.setLeft(formBox);
// Center - Table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
HBox searchBox = new HBox(10);
TextField searchField = new TextField();
searchField.setPromptText("Search by name...");
Button searchBtn = new Button("Search");
searchBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;");
searchBtn.setOnAction(e -> {
String q = searchField.getText().trim();
if (q.isEmpty()) refreshTable();
else {
tableData = FXCollections.observableArrayList(customerService.searchByName(q));
tableView.setItems(tableData);
```

```
}
});
Button showAllBtn = new Button("Show All");
showAllBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
showAllBtn.setOnAction(e -> refreshTable());
searchBox.getChildren().addAll(searchField, searchBtn, showAllBtn);
tableView = new TableView<>();
TableColumn<Customer, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("customerId"));
TableColumn<Customer, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(150);
TableColumn<Customer, String> phoneCol = new TableColumn<>("Phone");
phoneCol.setCellValueFactory(new PropertyValueFactory<>("phone"));
phoneCol.setPrefWidth(120);
TableColumn<Customer, String> emailCol = new TableColumn<>("Email");
emailCol.setCellValueFactory(new PropertyValueFactory<>("email"));
emailCol.setPrefWidth(170);
TableColumn<Customer, String> addrCol = new TableColumn<>("Address");
addrCol.setCellValueFactory(new PropertyValueFactory<>("address"));
addrCol.setPrefWidth(180);
tableView.getColumns().addAll(idCol, nameCol, phoneCol, emailCol, addrCol);
refreshTable();
tableView.getSelectionModel().selectedItemProperty().addListener((obs, o, n) -> {
if (n != null) {
idField.setText(n.getCustomerId());
nameField.setText(n.getName());
phoneField.setText(n.getPhone());
emailField.setText(n.getEmail());
addressField.setText(n.getAddress());
}
});
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(searchBox, tableView);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(customerService.getAllCustomers());
tableView.setItems(tableData);
}
private void clearFields(TextField... fields) {
for (TextField f : fields) f.clear();
tableView.getSelectionModel().clearSelection();
}
public Scene getScene() { return scene; }
}
```

28. PrescriptionScreen.java

Prescription handling screen. Displays prescriptions in a ListView and shows details in a TextArea. Allows creating new prescriptions with medicine items.

Code for PrescriptionScreen.java:

```
package com.pharmacy.ui;
```

```
import com.pharmacy.model.*;
import com.pharmacy.service.*;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.text.SimpleDateFormat;
import java.util.Date;
/**
* Prescription Handling Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, ArrayList operations.
*/
public class PrescriptionScreen {
private Scene scene;
private PharmacyApp app;
private PrescriptionService prescriptionService;
private CustomerService customerService;
private InventoryManager inventoryManager;
private TableView<Prescription> tableView;
private ObservableList<Prescription> tableData;
public PrescriptionScreen(PharmacyApp app, PrescriptionService prescriptionService,
CustomerService customerService, InventoryManager inventoryManager) {
this.app = app;
this.prescriptionService = prescriptionService;
this.customerService = customerService;
this.inventoryManager = inventoryManager;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Prescription Handling");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Form
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
```

```
formBox.setPrefWidth(330);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("New Prescription");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField custIdField = new TextField();
custIdField.setPromptText("Customer ID");
TextField custNameField = new TextField();
custNameField.setPromptText("Patient name");
TextField doctorField = new TextField();
doctorField.setPromptText("Doctor name");
TextField notesField = new TextField();
notesField.setPromptText("Notes/Instructions");
custIdField.textProperty().addListener((obs, o, n) -> {
if (n != null && !n.isEmpty()) {
Customer cust = customerService.getCustomerById(n.trim());
if (cust != null) custNameField.setText(cust.getName());
}
});
int r = 0;
grid.add(new Label("Customer ID:"), 0, r); grid.add(custIdField, 1, r++);
grid.add(new Label("Patient Name:"), 0, r); grid.add(custNameField, 1, r++);
grid.add(new Label("Doctor:"), 0, r); grid.add(doctorField, 1, r++);
grid.add(new Label("Notes:"), 0, r); grid.add(notesField, 1, r++);
// Prescription items
Label itemsTitle = new Label("Prescribed Medicines");
itemsTitle.setFont(Font.font("Arial", FontWeight.BOLD, 13));
TextField medIdField = new TextField();
medIdField.setPromptText("Medicine ID");
Label medNameLabel = new Label("-");
TextField medQtyField = new TextField();
medQtyField.setPromptText("Quantity");
TextField dosageField = new TextField();
dosageField.setPromptText("e.g., 1-0-1");
medIdField.textProperty().addListener((obs, o, n) -> {
if (n != null && !n.isEmpty()) {
Medicine med = inventoryManager.getMedicineById(n.trim());
if (med != null) medNameLabel.setText(med.getName());
}
});
GridPane itemGrid = new GridPane();
itemGrid.setVgap(6);
itemGrid.setHgap(6);
int ir = 0;
itemGrid.add(new Label("Med ID:"), 0, ir); itemGrid.add(medIdField, 1, ir++);
itemGrid.add(new Label("Name:"), 0, ir); itemGrid.add(medNameLabel, 1, ir++);
itemGrid.add(new Label("Qty:"), 0, ir); itemGrid.add(medQtyField, 1, ir++);
itemGrid.add(new Label("Dosage:"), 0, ir); itemGrid.add(dosageField, 1, ir++);
// Items list display
ListView<String> itemsList = new ListView<>();
itemsList.setPrefHeight(120);
java.util.ArrayList<PrescriptionItem> presItems = new java.util.ArrayList<>();
```

```
Label statusLabel = new Label();
Button addItemBtn = new Button("Add Medicine");
addItemBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;
-fx-pref-width: 200;");
addItemBtn.setOnAction(e -> {
try {
String medId = medIdField.getText().trim();
Medicine med = inventoryManager.getMedicineById(medId);
if (med == null) { statusLabel.setText("Medicine not found!"); return; }
int qty = Integer.parseInt(medQtyField.getText().trim());
PrescriptionItem item = new PrescriptionItem(medId, med.getName(), qty, dosageField.getText());
presItems.add(item);
itemsList.getItems().add(item.toString());
medIdField.clear();
medQtyField.clear();
dosageField.clear();
medNameLabel.setText("-");
} catch (Exception ex) {
statusLabel.setText("Error: " + ex.getMessage());
}
});
Button saveBtn = new Button("Save Prescription");
saveBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-font-size: 13; " +
"-fx-pref-width: 280; -fx-cursor: hand;");
saveBtn.setOnAction(e -> {
if (custNameField.getText().trim().isEmpty() || doctorField.getText().trim().isEmpty()) {
statusLabel.setText("Fill all fields!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
String presId = IDGenerator.generatePrescriptionId();
String date = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
Prescription pres = new Prescription(presId, custIdField.getText().trim(),
custNameField.getText(), doctorField.getText(), date, notesField.getText());
for (PrescriptionItem item : presItems) {
pres.addItem(item);
}
prescriptionService.addPrescription(pres);
refreshTable();
presItems.clear();
itemsList.getItems().clear();
custIdField.clear(); custNameField.clear(); doctorField.clear(); notesField.clear();
statusLabel.setText("Prescription saved: " + presId);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
formBox.getChildren().addAll(formTitle, grid, new Separator(),
itemsTitle, itemGrid, addItemBtn, itemsList, new Separator(),
saveBtn, statusLabel);
root.setLeft(formBox);
// Center - Table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
Label tableTitle = new Label("All Prescriptions");
tableTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
tableView = new TableView<>();
TableColumn<Prescription, String> idCol = new TableColumn<>("ID");
```

```
idCol.setCellValueFactory(new PropertyValueFactory<>("prescriptionId"));
TableColumn<Prescription, String> patCol = new TableColumn<>("Patient");
patCol.setCellValueFactory(new PropertyValueFactory<>("customerName"));
patCol.setPrefWidth(130);
TableColumn<Prescription, String> docCol = new TableColumn<>("Doctor");
docCol.setCellValueFactory(new PropertyValueFactory<>("doctorName"));
docCol.setPrefWidth(130);
TableColumn<Prescription, String> dateCol = new TableColumn<>("Date");
dateCol.setCellValueFactory(new PropertyValueFactory<>("date"));
TableColumn<Prescription, String> notesCol = new TableColumn<>("Notes");
notesCol.setCellValueFactory(new PropertyValueFactory<>("notes"));
notesCol.setPrefWidth(180);
tableView.getColumns().addAll(idCol, patCol, docCol, dateCol, notesCol);
refreshTable();
// Details area
TextArea detailsArea = new TextArea();
detailsArea.setEditable(false);
detailsArea.setPrefHeight(150);
detailsArea.setPromptText("Select a prescription to view details...");
tableView.getSelectionModel().selectedItemProperty().addListener((obs, o, n) -> {
if (n != null) {
StringBuilder sb = new StringBuilder();
sb.append("Prescription: ").append(n.getPrescriptionId()).append("\n");
sb.append("Patient: ").append(n.getCustomerName()).append("\n");
sb.append("Doctor: Dr. ").append(n.getDoctorName()).append("\n");
sb.append("Date: ").append(n.getDate()).append("\n");
sb.append("Notes: ").append(n.getNotes()).append("\n\n");
sb.append("Prescribed Medicines:\n");
for (PrescriptionItem item : n.getItems()) {
sb.append(" - ").append(item.toString()).append("\n");
}
detailsArea.setText(sb.toString());
}
});
Button deleteBtn = new Button("Delete Selected");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
deleteBtn.setOnAction(e -> {
Prescription sel = tableView.getSelectionModel().getSelectedItem();
if (sel != null) {
prescriptionService.deletePrescription(sel.getPrescriptionId());
refreshTable();
detailsArea.clear();
}
});
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(tableTitle, tableView, new Label("Details:"), detailsArea,
deleteBtn);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(prescriptionService.getAllPrescriptions());
tableView.setItems(tableData);
}
public Scene getScene() { return scene; }
```

```
}
```

29. ReportsScreen.java

Reports and backup screen. Shows summary statistics (medicine count, customer count, bill count) as cards. Provides a Backup button that triggers BackupService in a new Thread. Also shows low stock and expiry alerts.

Code for ReportsScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Bill;
import com.pharmacy.model.Medicine;
import com.pharmacy.service.*;
import javafx.application.Platform;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.Date;
/**
* Reports & Alerts Screen - Sales reports, stock alerts, expiry alerts, backups.
* Demonstrates: JavaFX TableView, VBox/HBox, Multithreading integration, Backup service.
*/
public class ReportsScreen {
private Scene scene;
private PharmacyApp app;
private InventoryManager inventoryManager;
private BillingService billingService;
private CustomerService customerService;
private SupplierService supplierService;
private ExpiryAlertThread expiryAlertThread;
private StockAlertThread stockAlertRunnable;
public ReportsScreen(PharmacyApp app, InventoryManager inventoryManager,
BillingService billingService, CustomerService customerService,
SupplierService supplierService,
ExpiryAlertThread expiryAlertThread, StockAlertThread stockAlertRunnable) {
this.app = app;
this.inventoryManager = inventoryManager;
this.billingService = billingService;
this.customerService = customerService;
this.supplierService = supplierService;
this.expiryAlertThread = expiryAlertThread;
this.stockAlertRunnable = stockAlertRunnable;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
```

```
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Reports & Alerts");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Report categories
VBox leftMenu = new VBox(8);
leftMenu.setPadding(new Insets(15));
leftMenu.setPrefWidth(200);
leftMenu.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0
1 0 0;");
Label menuTitle = new Label("Report Types");
menuTitle.setFont(Font.font("Arial", FontWeight.BOLD, 14));
// Content area in center
VBox contentArea = new VBox(10);
contentArea.setPadding(new Insets(15));
Button salesReportBtn = createMenuButton("Sales Report", "#27ae60");
Button stockReportBtn = createMenuButton("Stock Report", "#3498db");
Button expiryAlertBtn = createMenuButton("Expiry Alerts", "#e74c3c");
Button lowStockBtn = createMenuButton("Low Stock Alert", "#e67e22");
Button summaryBtn = createMenuButton("Summary", "#9b59b6");
Button backupBtn = createMenuButton("Backup Data", "#34495e");
// Sales Report
salesReportBtn.setOnAction(e -> showSalesReport(contentArea));
// Stock Report
stockReportBtn.setOnAction(e -> showStockReport(contentArea));
// Expiry Alerts (from alert thread)
expiryAlertBtn.setOnAction(e -> showExpiryAlerts(contentArea));
// Low Stock Alert (from stock thread)
lowStockBtn.setOnAction(e -> showLowStockAlerts(contentArea));
// Summary
summaryBtn.setOnAction(e -> showSummary(contentArea));
// Backup
backupBtn.setOnAction(e -> runBackup(contentArea));
leftMenu.getChildren().addAll(menuTitle, new Separator(),
salesReportBtn, stockReportBtn, expiryAlertBtn,
lowStockBtn, summaryBtn, new Separator(), backupBtn);
root.setLeft(leftMenu);
// Initial content
showSummary(contentArea);
root.setCenter(contentArea);
scene = new Scene(root, 1100, 700);
}
private void showSalesReport(VBox contentArea) {
```

```
contentArea.getChildren().clear();
Label title = new Label("Sales Report");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
String today = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
Label totalSales = new Label(String.format("Total Sales (All Time): Rs.%.2f",
billingService.getTotalSales()));
totalSales.setFont(Font.font("Arial", FontWeight.BOLD, 14));
totalSales.setStyle("-fx-text-fill: #27ae60;");
Label todaySales = new Label(String.format("Today's Sales: Rs.%.2f",
billingService.getTotalSalesByDate(today)));
todaySales.setFont(Font.font("Arial", FontWeight.BOLD, 14));
Label billCount = new Label("Total Bills: " + billingService.getBillCount());
// Bills table
TableView<Bill> billTable = new TableView<>();
ObservableList<Bill> bills = FXCollections.observableArrayList(billingService.getAllBills());
billTable.setItems(bills);
TableColumn<Bill, String> idCol = new TableColumn<>("Bill ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("billId"));
TableColumn<Bill, String> custCol = new TableColumn<>("Customer");
custCol.setCellValueFactory(new PropertyValueFactory<>("customerName"));
custCol.setPrefWidth(150);
TableColumn<Bill, String> dateCol = new TableColumn<>("Date");
dateCol.setCellValueFactory(new PropertyValueFactory<>("date"));
TableColumn<Bill, Double> totalCol = new TableColumn<>("Total");
totalCol.setCellValueFactory(new PropertyValueFactory<>("totalAmount"));
TableColumn<Bill, Double> discCol = new TableColumn<>("Discount");
discCol.setCellValueFactory(new PropertyValueFactory<>("discount"));
TableColumn<Bill, Double> netCol = new TableColumn<>("Net Amount");
netCol.setCellValueFactory(new PropertyValueFactory<>("netAmount"));
netCol.setPrefWidth(100);
TableColumn<Bill, String> byCol = new TableColumn<>("Billed By");
byCol.setCellValueFactory(new PropertyValueFactory<>("billedBy"));
billTable.getColumns().addAll(idCol, custCol, dateCol, totalCol, discCol, netCol, byCol);
VBox.setVgrow(billTable, Priority.ALWAYS);
contentArea.getChildren().addAll(title, totalSales, todaySales, billCount, billTable);
}
@SuppressWarnings("unchecked")
private void showStockReport(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Stock Report");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
Label countLabel = new Label("Total Medicines in Inventory: " +
inventoryManager.getMedicineCount());
countLabel.setFont(Font.font("Arial", FontWeight.BOLD, 14));
TableView<Medicine> table = new TableView<>();
ObservableList<Medicine> data =
FXCollections.observableArrayList(inventoryManager.getSortedByName());
table.setItems(data);
TableColumn<Medicine, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("medicineId"));
TableColumn<Medicine, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(170);
TableColumn<Medicine, String> catCol = new TableColumn<>("Category");
catCol.setCellValueFactory(new PropertyValueFactory<>("category"));
```

```
TableColumn<Medicine, Integer> qtyCol = new TableColumn<>("Quantity");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
TableColumn<Medicine, Double> priceCol = new TableColumn<>("Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("price"));
TableColumn<Medicine, String> expCol = new TableColumn<>("Expiry");
expCol.setCellValueFactory(new PropertyValueFactory<>("expiryDate"));
expCol.setPrefWidth(100);
table.getColumns().addAll(idCol, nameCol, catCol, qtyCol, priceCol, expCol);
VBox.setVgrow(table, Priority.ALWAYS);
contentArea.getChildren().addAll(title, countLabel, table);
}
@SuppressWarnings("unchecked")
private void showExpiryAlerts(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Expiry Alerts");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
title.setStyle("-fx-text-fill: #e74c3c;");
SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");
String currentDate = sdf.format(new Date());
Calendar cal = Calendar.getInstance();
cal.add(Calendar.DAY_OF_MONTH, 30);
String thresholdDate = sdf.format(cal.getTime());
// Expired medicines
ArrayList<Medicine> expired = inventoryManager.getExpiredMedicines(currentDate);
Label expiredTitle = new Label("EXPIRED Medicines (" + expired.size() + ")");
expiredTitle.setFont(Font.font("Arial", FontWeight.BOLD, 14));
expiredTitle.setStyle("-fx-text-fill: #e74c3c;");
TableView<Medicine> expiredTable = new TableView<>();
expiredTable.setItems(FXCollections.observableArrayList(expired));
addMedicineColumns(expiredTable);
expiredTable.setPrefHeight(200);
// Expiring soon
ArrayList<Medicine> expiringSoon = inventoryManager.getExpiringSoon(currentDate,
thresholdDate);
Label expiringTitle = new Label("Expiring Within 30 Days (" + expiringSoon.size() + ")");
expiringTitle.setFont(Font.font("Arial", FontWeight.BOLD, 14));
expiringTitle.setStyle("-fx-text-fill: #e67e22;");
TableView<Medicine> expiringTable = new TableView<>();
expiringTable.setItems(FXCollections.observableArrayList(expiringSoon));
addMedicineColumns(expiringTable);
VBox.setVgrow(expiringTable, Priority.ALWAYS);
contentArea.getChildren().addAll(title, expiredTitle, expiredTable, expiringTitle,
expiringTable);
}
@SuppressWarnings("unchecked")
private void showLowStockAlerts(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Low Stock Alerts (< 10 units)");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
title.setStyle("-fx-text-fill: #e67e22;");
ArrayList<Medicine> lowStock = inventoryManager.getLowStockMedicines();
Label countLabel = new Label("Low Stock Items: " + lowStock.size());
countLabel.setFont(Font.font("Arial", FontWeight.BOLD, 14));
TableView<Medicine> table = new TableView<>();
table.setItems(FXCollections.observableArrayList(lowStock));
```

```
addMedicineColumns(table);
VBox.setVgrow(table, Priority.ALWAYS);
contentArea.getChildren().addAll(title, countLabel, table);
}
private void showSummary(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("System Summary");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
GridPane summaryGrid = new GridPane();
summaryGrid.setVgap(15);
summaryGrid.setHgap(20);
summaryGrid.setPadding(new Insets(20));
addSummaryCard(summaryGrid, 0, 0, "Total Medicines",
String.valueOf(inventoryManager.getMedicineCount()), "#3498db");
addSummaryCard(summaryGrid, 1, 0, "Total Customers",
String.valueOf(customerService.getCustomerCount()), "#9b59b6");
addSummaryCard(summaryGrid, 2, 0, "Total Suppliers",
String.valueOf(supplierService.getSupplierCount()), "#e67e22");
addSummaryCard(summaryGrid, 0, 1, "Total Bills",
String.valueOf(billingService.getBillCount()), "#27ae60");
addSummaryCard(summaryGrid, 1, 1, "Total Sales",
String.format("Rs.%.2f", billingService.getTotalSales()), "#2c3e50");
String currentDate = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
addSummaryCard(summaryGrid, 2, 1, "Low Stock Items",
String.valueOf(inventoryManager.getLowStockMedicines().size()), "#e74c3c");
// Thread status
Label threadStatus = new Label("Background Threads:");
threadStatus.setFont(Font.font("Arial", FontWeight.BOLD, 14));
Label expiryThreadLabel = new Label("Expiry Alert Thread: " +
(expiryAlertThread != null && expiryAlertThread.isAlive() ? "RUNNING" : "STOPPED"));
expiryThreadLabel.setStyle(expiryAlertThread != null && expiryAlertThread.isAlive() ?
"-fx-text-fill: #27ae60;" : "-fx-text-fill: #e74c3c;");
Label stockThreadLabel = new Label("Stock Alert Thread: RUNNING (daemon)");
stockThreadLabel.setStyle("-fx-text-fill: #27ae60;");
contentArea.getChildren().addAll(title, summaryGrid, new Separator(),
threadStatus, expiryThreadLabel, stockThreadLabel);
}
private void addSummaryCard(GridPane grid, int col, int row, String label, String value, String
color) {
VBox card = new VBox(5);
card.setPadding(new Insets(15));
card.setPrefSize(180, 80);
card.setAlignment(Pos.CENTER);
card.setStyle("-fx-background-color: " + color + "; -fx-background-radius: 8;");
Label valLabel = new Label(value);
valLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
valLabel.setStyle("-fx-text-fill: white;");
Label nameLabel = new Label(label);
nameLabel.setStyle("-fx-text-fill: #ecf0f1; -fx-font-size: 12;");
card.getChildren().addAll(valLabel, nameLabel);
grid.add(card, col, row);
}
private void runBackup(VBox contentArea) {
contentArea.getChildren().clear();
```

```
Label title = new Label("Data Backup");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
Label statusLabel = new Label("Starting backup...");
statusLabel.setStyle("-fx-text-fill: #3498db; -fx-font-size: 14;");
ProgressIndicator progress = new ProgressIndicator();
progress.setPrefSize(50, 50);
contentArea.getChildren().addAll(title, progress, statusLabel);
// Run backup in a separate thread (BackupService implements Runnable)
BackupService backupService = new BackupService(
inventoryManager, customerService, supplierService,
billingService, app.getPrescriptionService());
backupService.setCallback(message -> {
Platform.runLater(() -> {
statusLabel.setText(message);
statusLabel.setStyle("-fx-text-fill: #27ae60; -fx-font-size: 14;");
progress.setProgress(1.0);
Label pathLabel = new Label("Backup files saved in: data/ directory");
pathLabel.setStyle("-fx-font-size: 12;");
Label filesLabel = new Label(
"Files created:\n" +
" - backup_medicines.txt\n" +
" - backup_customers.txt\n" +
" - backup_suppliers.txt\n" +
" - backup_bills.txt\n" +
" - backup_prescriptions.txt");
filesLabel.setStyle("-fx-font-size: 12;");
contentArea.getChildren().addAll(pathLabel, filesLabel);
});
});
Thread backupThread = new Thread(backupService, "BackupThread");
backupThread.setDaemon(true);
backupThread.start();
}
@SuppressWarnings("unchecked")
private void addMedicineColumns(TableView<Medicine> table) {
TableColumn<Medicine, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("medicineId"));
TableColumn<Medicine, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(150);
TableColumn<Medicine, String> catCol = new TableColumn<>("Category");
catCol.setCellValueFactory(new PropertyValueFactory<>("category"));
TableColumn<Medicine, Integer> qtyCol = new TableColumn<>("Qty");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
TableColumn<Medicine, Double> priceCol = new TableColumn<>("Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("price"));
TableColumn<Medicine, String> expCol = new TableColumn<>("Expiry");
expCol.setCellValueFactory(new PropertyValueFactory<>("expiryDate"));
expCol.setPrefWidth(100);
table.getColumns().addAll(idCol, nameCol, catCol, qtyCol, priceCol, expCol);
}
private Button createMenuButton(String text, String color) {
Button btn = new Button(text);
btn.setPrefWidth(170);
btn.setStyle("-fx-background-color: " + color + "; -fx-text-fill: white; " +
```

```
"-fx-font-size: 12; -fx-cursor: hand; -fx-alignment: center-left;");
return btn;
}
public Scene getScene() { return scene; }
}
```

Util Package (com.pharmacy.util)

30. DataStore.java

Generic utility class DataStore<T extends Serializable> that handles all persistence. saveAll() serializes an ArrayList using ObjectOutputStream + FileOutputStream. loadAll() deserializes using ObjectInputStream + FileInputStream. backupToText() writes readable text using FileWriter. readBackup() reads with FileReader. All methods are synchronized.

Code for DataStore.java:

```
package com.pharmacy.util;
import java.io.*;
import java.util.ArrayList;
/**
* Generic data store for serialization and deserialization.
* Demonstrates: Generics, Serialization/Deserialization, File I/O.
*
* @param <T> The type of objects to store (must be Serializable)
*/
public class DataStore<T extends Serializable> {
private String filePath;
public DataStore(String filePath) {
this.filePath = filePath;
}
/**
* Save a list of objects using Serialization (ObjectOutputStream).
* Demonstrates: FileOutputStream, ObjectOutputStream, Serialization.
*/
public synchronized void saveAll(ArrayList<T> items) {
try (FileOutputStream fos = new FileOutputStream(filePath);
ObjectOutputStream oos = new ObjectOutputStream(fos)) {
oos.writeObject(items);
} catch (IOException e) {
System.err.println("Error saving data to " + filePath + ": " + e.getMessage());
}
}
/**
* Load a list of objects using Deserialization (ObjectInputStream).
* Demonstrates: FileInputStream, ObjectInputStream, Deserialization.
*/
@SuppressWarnings("unchecked")
public synchronized ArrayList<T> loadAll() {
File file = new File(filePath);
if (!file.exists()) {
return new ArrayList<>();
}
try (FileInputStream fis = new FileInputStream(filePath);
ObjectInputStream ois = new ObjectInputStream(fis)) {
return (ArrayList<T>) ois.readObject();
```

```
} catch (IOException | ClassNotFoundException e) {
System.err.println("Error loading data from " + filePath + ": " + e.getMessage());
return new ArrayList<>();
}
}
/**
* Backup data to a text file using FileWriter.
* Demonstrates: FileWriter usage.
*/
public void backupToText(ArrayList<T> items, String backupPath) {
try (FileWriter writer = new FileWriter(backupPath)) {
writer.write("=== DATA BACKUP ===\n");
writer.write("Source: " + filePath + "\n");
writer.write("Records: " + items.size() + "\n");
writer.write("====================\n\n");
for (T item : items) {
writer.write(item.toString() + "\n");
}
writer.write("\n=== END OF BACKUP ===\n");
} catch (IOException e) {
System.err.println("Error creating backup: " + e.getMessage());
}
}
/**
* Read backup file content using FileReader.
* Demonstrates: FileReader usage.
*/
public String readBackup(String backupPath) {
StringBuilder content = new StringBuilder();
File file = new File(backupPath);
if (!file.exists()) {
return "No backup found.";
}
try (FileReader reader = new FileReader(backupPath)) {
int ch;
while ((ch = reader.read()) != -1) {
content.append((char) ch);
}
} catch (IOException e) {
System.err.println("Error reading backup: " + e.getMessage());
}
return content.toString();
}
public String getFilePath() {
return filePath;
}
}
```

31. IDGenerator.java

Utility class with synchronized static methods for generating unique IDs. Generates IDs in the format MED-001, CUS-001, SUP-001, BIL-001, PRE-001 using an internal counter for each type.

Code for IDGenerator.java:

```
package com.pharmacy.util;
/**
```

```
* Utility class for generating unique IDs.
* Demonstrates: Static methods, synchronized for thread safety.
*/
public class IDGenerator {
private static int medicineCounter = 1000;
private static int customerCounter = 2000;
private static int supplierCounter = 3000;
private static int billCounter = 4000;
private static int prescriptionCounter = 5000;
// Synchronized to ensure thread-safe ID generation
public static synchronized String generateMedicineId() {
return "MED" + (++medicineCounter);
}
public static synchronized String generateCustomerId() {
return "CUS" + (++customerCounter);
}
public static synchronized String generateSupplierId() {
return "SUP" + (++supplierCounter);
}
public static synchronized String generateBillId() {
return "BIL" + (++billCounter);
}
public static synchronized String generatePrescriptionId() {
return "PRE" + (++prescriptionCounter);
}
// Update counters based on existing data
public static void updateCounters(int medCount, int cusCount, int supCount,
int bilCount, int preCount) {
if (medCount > medicineCounter) medicineCounter = medCount;
if (cusCount > customerCounter) customerCounter = cusCount;
if (supCount > supplierCounter) supplierCounter = supCount;
if (bilCount > billCounter) billCounter = bilCount;
if (preCount > prescriptionCounter) prescriptionCounter = preCount;
}
}
```

32. SampleDataLoader.java

Utility class that loads sample data on the first run of the application. Adds 10 medicines, 3 customers, and 3 suppliers so the system has data to demonstrate.

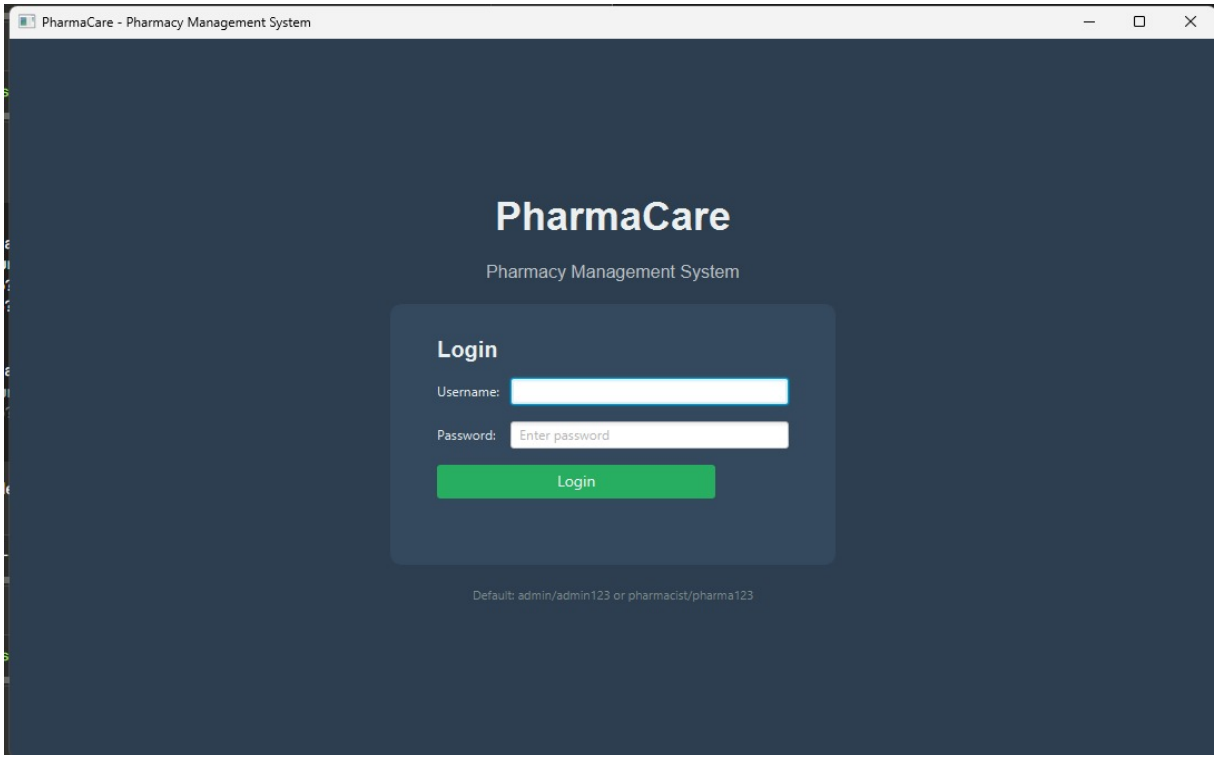
Code for SampleDataLoader.java:

```
package com.pharmacy.util;
import com.pharmacy.model.*;
import com.pharmacy.service.*;
/**
* Loads sample data for demonstration purposes.
* Demonstrates: Object creation, Enum usage, ArrayList operations.
*/
public class SampleDataLoader {
public static void loadSampleData(InventoryManager inventoryManager,
CustomerService customerService,
SupplierService supplierService) {
// Only load if no data exists
if (inventoryManager.getMedicineCount() > 0) {
```

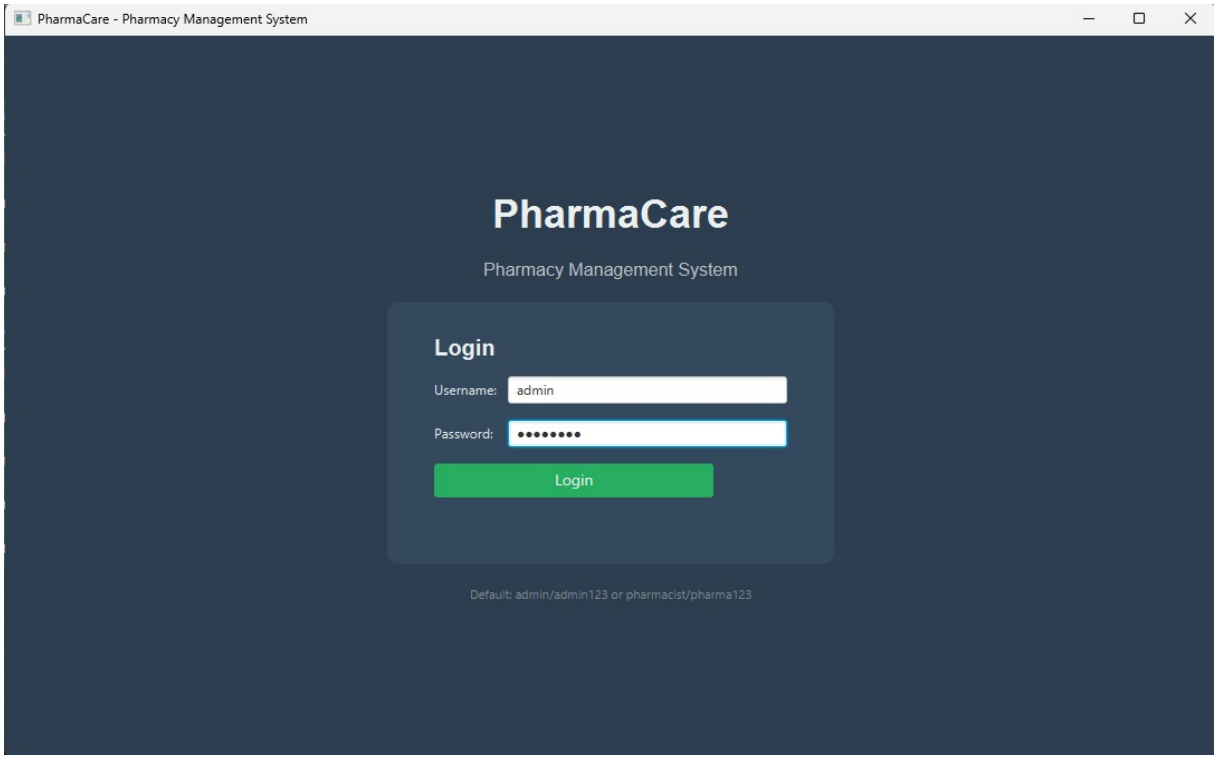
```
return;
}
// Sample Suppliers
Supplier s1 = new Supplier("SUP3001", "PharmaCorp India", "Rajesh Kumar",
"9876543210", "rajesh@pharmacorp.com", "Mumbai, Maharashtra");
Supplier s2 = new Supplier("SUP3002", "MediLife Supplies", "Priya Sharma",
"9876543211", "priya@medilife.com", "Delhi, India");
Supplier s3 = new Supplier("SUP3003", "HealthFirst Pharma", "Amit Patel",
"9876543212", "amit@healthfirst.com", "Ahmedabad, Gujarat");
supplierService.addSupplier(s1);
supplierService.addSupplier(s2);
supplierService.addSupplier(s3);
// Sample Medicines
inventoryManager.addMedicine(new Medicine("MED1001", "Paracetamol 500mg", "Sun Pharma",
MedicineCategory.TABLET, "BT001", "2026-12-15", 500, 12.50, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1002", "Amoxicillin 250mg", "Cipla",
MedicineCategory.CAPSULE, "BT002", "2026-06-20", 200, 45.00, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1003", "Cough Syrup", "Himalaya",
MedicineCategory.SYRUP, "BT003", "2026-03-10", 150, 85.00, "SUP3002"));
inventoryManager.addMedicine(new Medicine("MED1004", "Insulin Injection", "Novo Nordisk",
MedicineCategory.INJECTION, "BT004", "2025-08-25", 50, 350.00, "SUP3003"));
inventoryManager.addMedicine(new Medicine("MED1005", "Dettol Ointment", "Reckitt",
MedicineCategory.OINTMENT, "BT005", "2027-01-30", 300, 65.00, "SUP3002"));
inventoryManager.addMedicine(new Medicine("MED1006", "Eye Drops", "Alcon",
MedicineCategory.DROPS, "BT006", "2026-09-15", 8, 120.00, "SUP3003"));
inventoryManager.addMedicine(new Medicine("MED1007", "Salbutamol Inhaler", "Cipla",
MedicineCategory.INHALER, "BT007", "2026-11-20", 5, 250.00, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1008", "ORS Powder", "WHO Standard",
MedicineCategory.POWDER, "BT008", "2027-06-10", 1000, 15.00, "SUP3002"));
inventoryManager.addMedicine(new Medicine("MED1009", "Azithromycin 500mg", "Zydus",
MedicineCategory.TABLET, "BT009", "2026-04-05", 100, 75.00, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1010", "Vitamin D3", "HealthVit",
MedicineCategory.CAPSULE, "BT010", "2026-10-30", 400, 30.00, "SUP3003"));
// Sample Customers
customerService.addCustomer(new Customer("CUS2001", "Rahul Verma", "9123456789",
"rahul@email.com", "Bangalore, Karnataka"));
customerService.addCustomer(new Customer("CUS2002", "Sneha Iyer", "9123456790",
"sneha@email.com", "Chennai, Tamil Nadu"));
customerService.addCustomer(new Customer("CUS2003", "Mohammed Ali", "9123456791",
"ali@email.com", "Hyderabad, Telangana"));
// Update ID counters
IDGenerator.updateCounters(1010, 2003, 3003, 4000, 5000);
}
}
```

Results

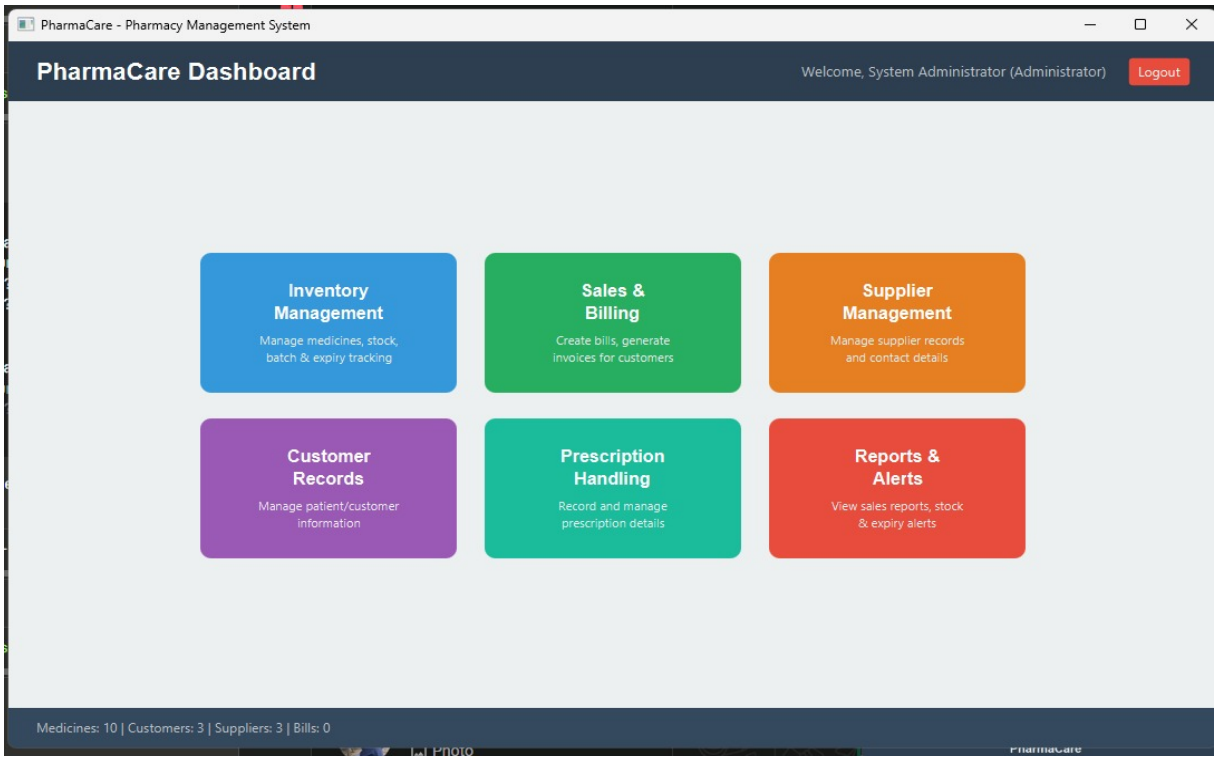
Login Screen



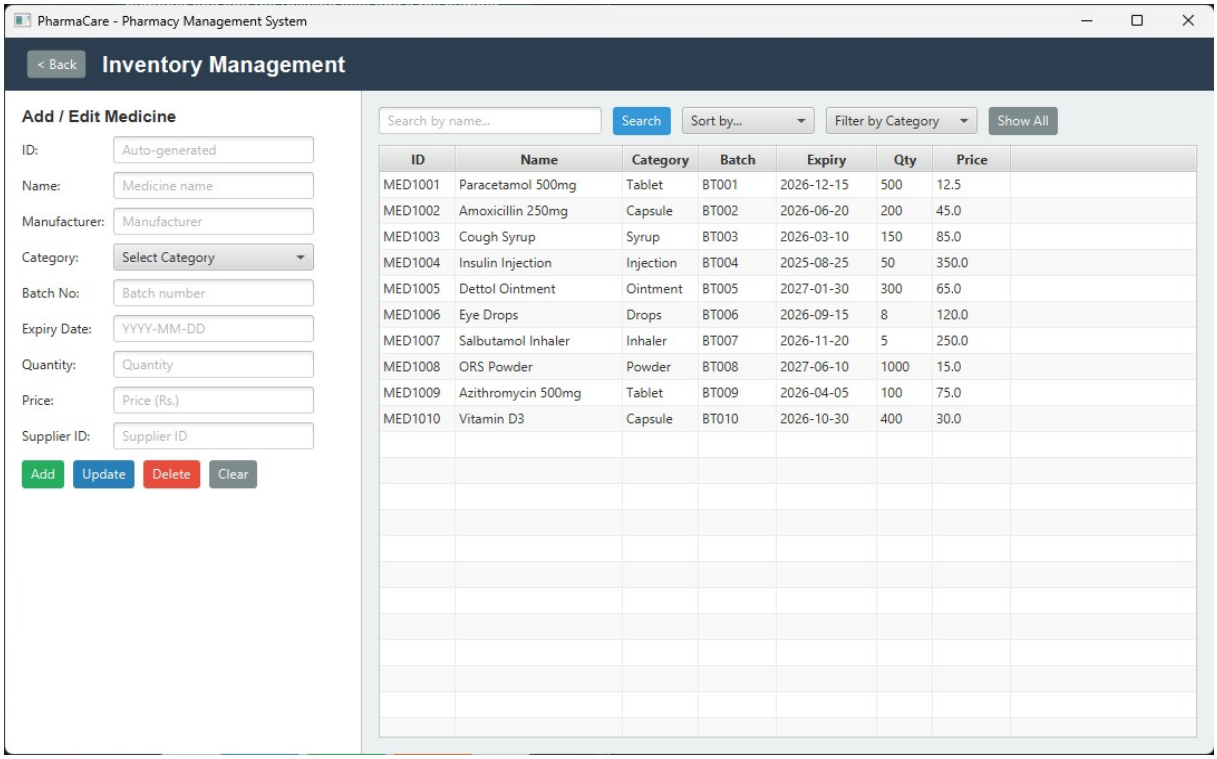
Login Screen - with Credentials



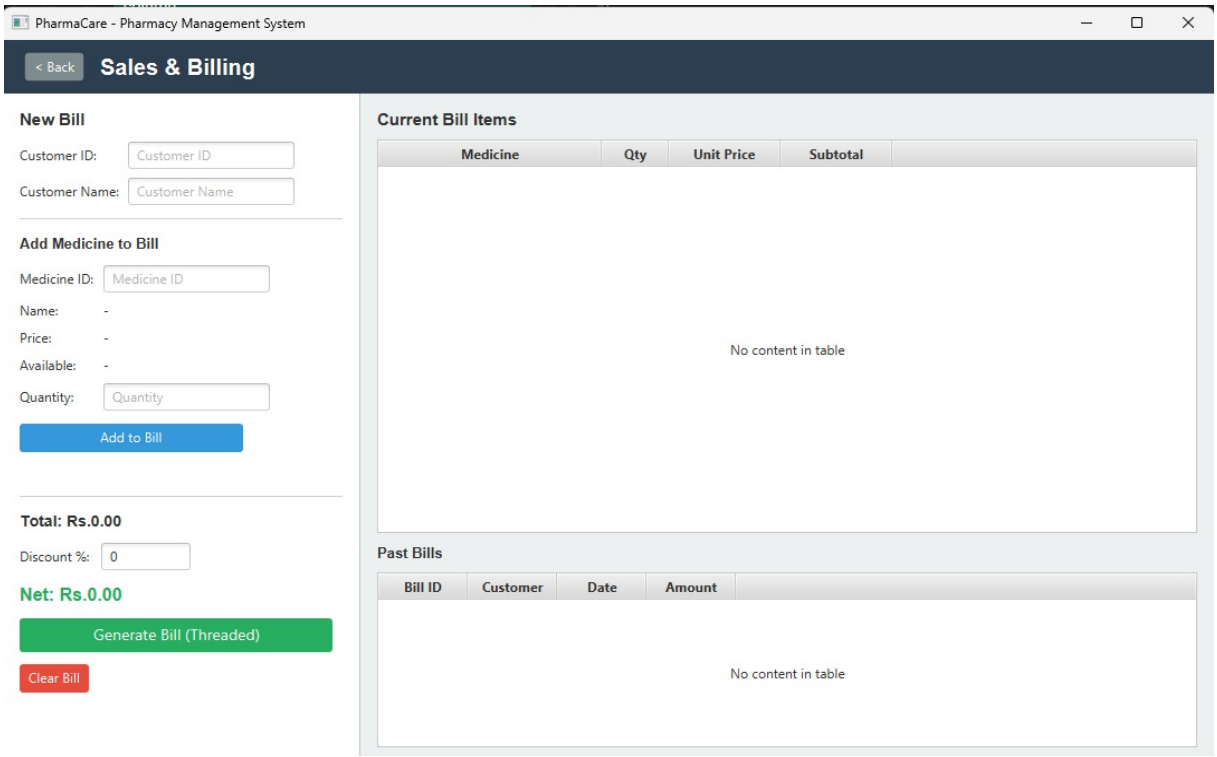
Dashboard Screen



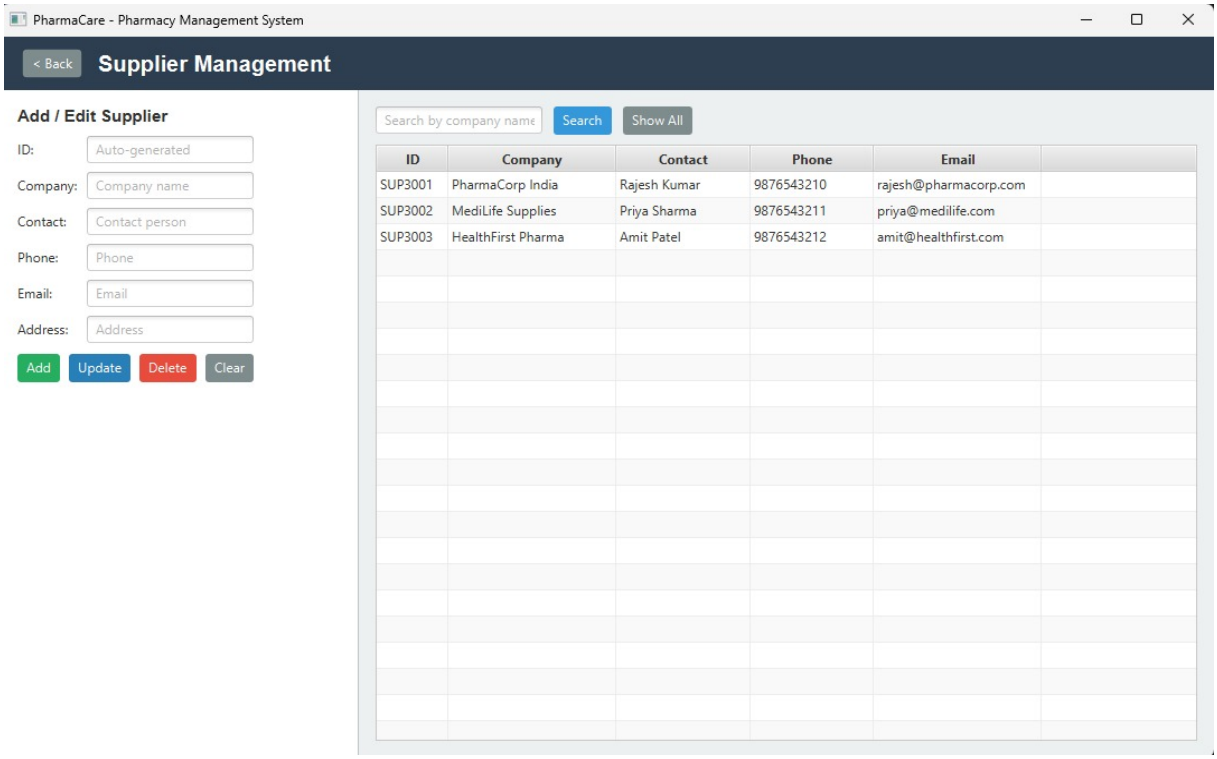
Inventory Management Screen



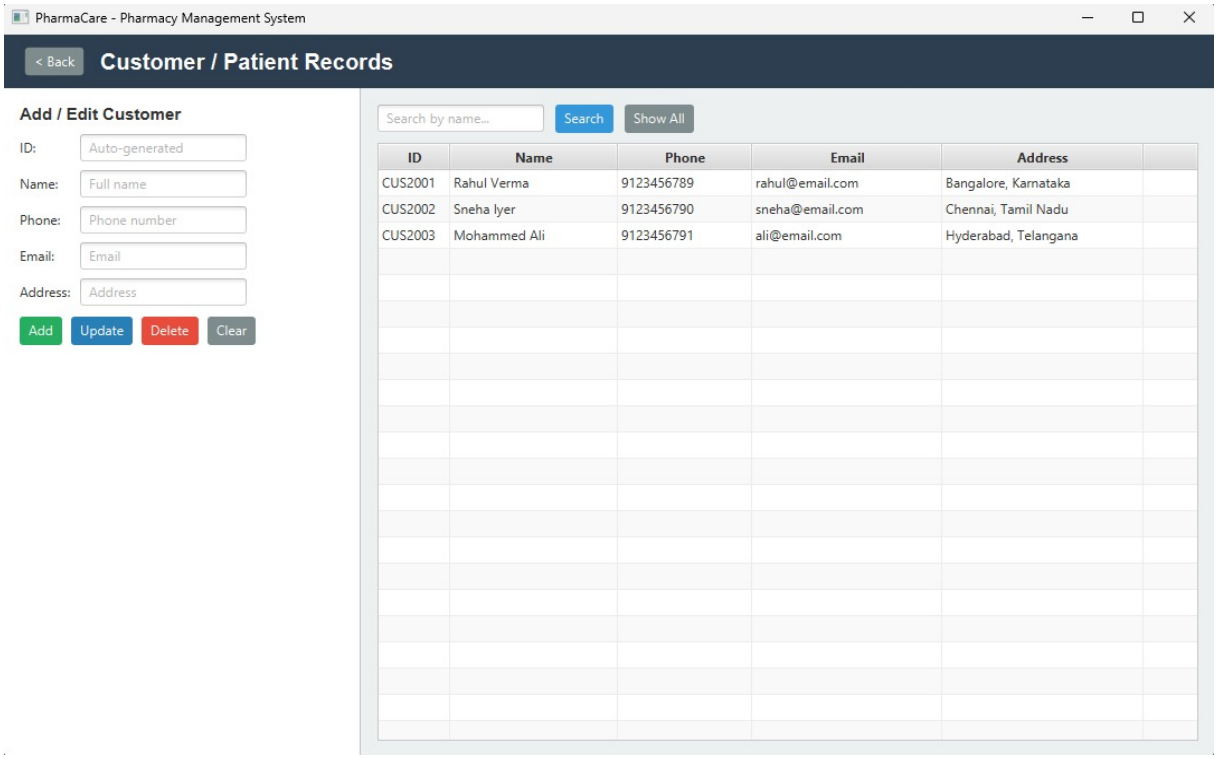
Sales and Billing Screen



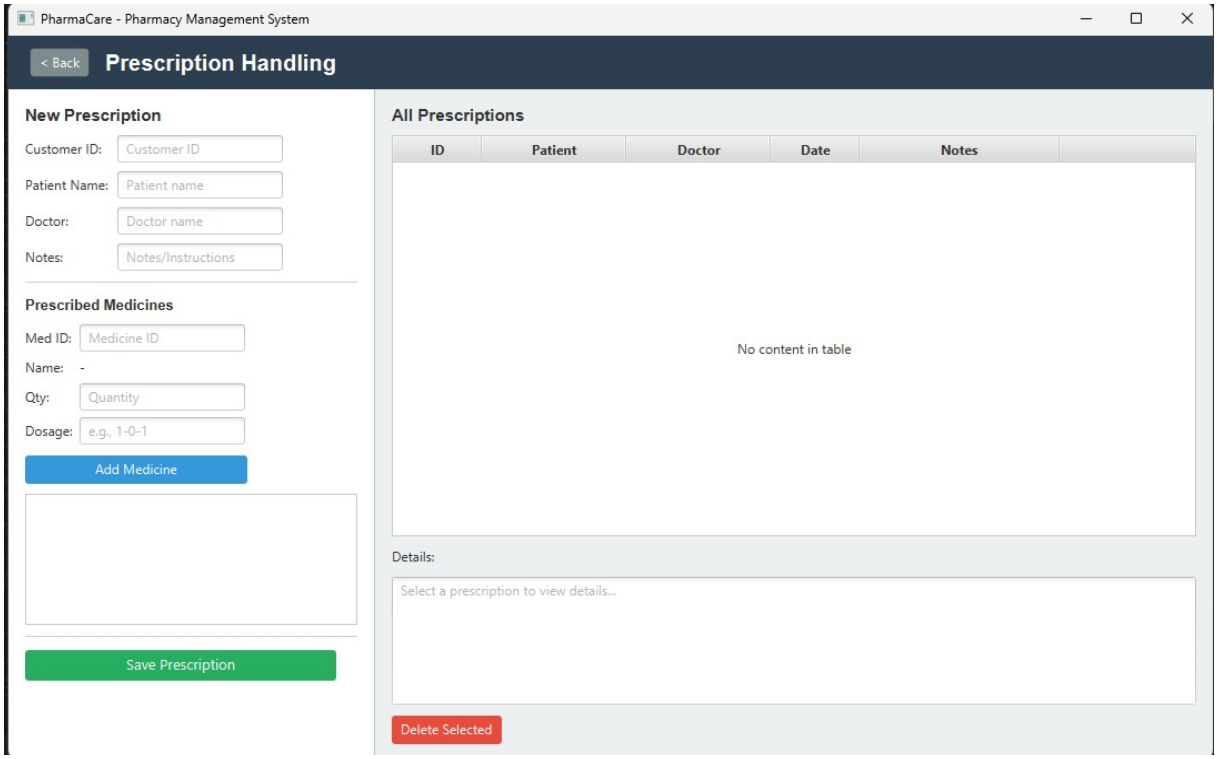
Supplier Management Screen



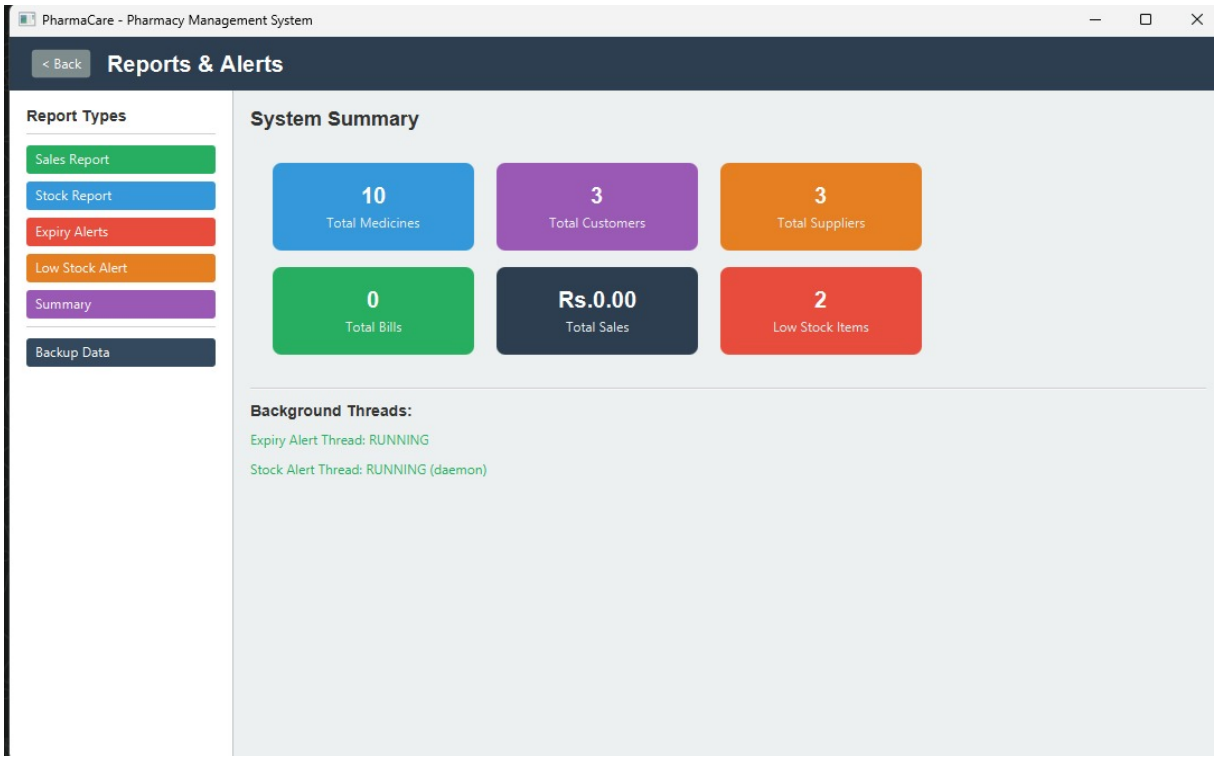
Customer Records Screen



Prescription Handling Screen



Reports and Alerts Screen


PharmaCare - Pharmacy Management System
< Back
Reports & Alerts
Report Types
Sales Report
Stock Report
Expiry Alerts
Low Stock Alert
Summary
Backup Data
System Summary
10
Total Medicines
3
Total Customers
3
Total Suppliers
0
Total Bills
Rs.0.00
Total Sales
2
Low Stock Items
Background Threads:
Expiry Alert Thread: RUNNING
Stock Alert Thread: RUNNING (daemon)

Conclusion

The Pharmacy Management System (PMS) provides an efficient and reliable platform for pharmacy staff to manage medicine inventory, process billing transactions, maintain customer and supplier records, handle prescriptions, and generate automated alerts for expiring or low-stock medicines. Using Java and JavaFX, the system delivers a clean, intuitive graphical interface that makes it easy for both administrators and pharmacists to interact with their data securely and accurately.

By integrating role-based access control, unique auto-generated IDs for each record, and synchronized thread-safe service methods, the PMS ensures that data integrity is preserved even under concurrent access conditions. The use of background threads for expiry and stock monitoring, combined with automatic invoice generation using FileWriter, makes the system both proactive and practical for day-to-day pharmacy operations.

Overall, the PMS successfully demonstrates how object-oriented programming principles - Encapsulation, Inheritance, Polymorphism, and Abstraction - along with Java Collections, Generics, Multithreading, File Handling, and Serialization can be applied to solve real-world problems in healthcare management, resulting in a secure, modular, and scalable solution.

Individual Contributions

Kenneth Martin, 240962047, Roll No. 8

- PharmacyApp.java
- LoginScreen.java
- DashboardScreen.java
- UserService.java
- User.java / UserRole.java

Sam Thomas, 240962242, Roll No. 18

- Medicine.java / MedicineCategory.java
- InventoryManager.java
- InventoryScreen.java
- SampleDataLoader.java

Mohammed Rayyan Sheik, 240962073, Roll No. 12

- BillingService.java
- BillingThread.java
- BillingScreen.java
- Bill.java / BillItem.java

Vedank Singh, 240962043, Roll No. 7

- DataStore.java
- IDGenerator.java
- CustomerService.java / SupplierService.java
- PrescriptionService.java
- ExpiryAlertThread.java / StockAlertThread.java
- BackupService.java
- ReportsScreen.java
- Supplier.java / Customer.java / Prescription.java / PrescriptionItem.java / BillItem.java

While project tasks were divided among the team members, we collaborated closely at every stage, discussing and refining our design decisions together. Since this system strictly follows Object-Oriented Programming principles, each file is interdependent, requiring continuous communication and coordination to ensure seamless integration across all classes and modules. By regularly reviewing each other's work and resolving challenges collaboratively, we maintained consistency in design and functionality, resulting in a unified and robust Pharmacy Management System.

References

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- 2. JavaFX Tutorials and Codes: https://www.geeksforgeeks.org/tag/javafx/
- 3. Java Serialization Guide: https://www.baeldung.com/java-serialization
- 4. Java Collections Framework: https://docs.oracle.com/javase/8/docs/technotes/guides/collections/overview.html
- 5. Java Multithreading: https://www.geeksforgeeks.org/multithreading-in-java/